CHAPTER 21

THE LEGENDRE POLYNOMIALS $P_n(x)$

Legendre polynomial functions provide solutions to differential equations that govern important physical phenomena. Named for the prolific French mathematician Adrien Marie Legendre (1752 – 1833), these polynomials constitute a family that is one of the simplest that exhibits orthogonality, a property described in Section 21:14.

21:1 NOTATION

The symbol $P_n(x)$ is standard for the Legendre polynomial of degree n and argument x, though the P is often italicized. The name *spherical polynomial* is also encountered, *zonal surface harmonic function* [Section 59:14] being yet another name. When orthogonality is important, *normalized* or *orthonormal* Legendre polynomials are often specified; this implies multiplication of $P_n(x)$ by $\sqrt{n+\frac{1}{2}}$, but some authors do not make the presence of this factor explicit.

The symbol $P_\nu(x)$, where ν is unrestricted, is given to a class of functions discussed in Chapter 59 and named *Legendre functions of the first kind*. The degree of a Legendre *polynomial*, on the other hand, is usually restricted to nonnegative integers. The two P functions are identical for integer ν of either sign because

21:1:1
$$P_\nu(x) = \begin{cases} P_n(x) & \nu = 0,1,2,\cdots \qquad n = \nu \\ P_{-n-1}(x) & \nu = -1,-2,-3\cdots \qquad n = -1-\nu \end{cases}$$

Thus the functions of this chapter are just special instances of Legendre functions of the first kind and are sometimes not regarded as distinct from the more general function.

Provided that it does not exceed unity in magnitude, the argument x of a Legendre polynomial is often replaced by the cosine of a subsidiary variable

21:1:2
$$P_n(\cos(\theta)) = P_n(x) \qquad \theta = \arccos(x) \qquad -1 \le x \le 1$$

This reflects the manner in which Legendre polynomials often arise in scientific and engineering applications.

The *shifted Legendre polynomials*, distinguished by an asterisk, have a changed argument

21:1:3
$$P_n^*(x) = P_n(2x-1)$$

so that the orthogonality interval [Section 21:14] becomes 0 to 1 instead of −1 to 1. Another modified Legendre polynomial is

K.B. Oldham et al., *An Atlas of Functions, Second Edition*,
DOI 10.1007/978-0-387-48807-3_22,

21:1:4 $$P^n(x) = \frac{(2n)!}{2^n} P_n(x)$$

Neither of these modifications is used in the *Atlas*. See Sections 21:12 and 22:12, respectively, for explanations of the $P_\nu^{(\mu)}(x)$ and $P_n^{(\alpha,\beta)}(x)$ notations.

21:2 BEHAVIOR

We allow the argument x to adopt any real value in this chapter, though its most important domain is $-1 \le x \le 1$ and restriction to this domain is obligatory when the argument is replaced by $\cos(\theta)$. As Figure 21-1 illustrates, the range of the Legendre polynomial reflects its argument and degree as follows

21:2:1 $$\begin{cases} 1 \le P_n(x) < \infty & x \ge 1, & n = 0,1,2,\cdots \\ -1 \le P_n(x) < 1 & -1 \le x \le 1, & n = 0,1,2,\cdots \\ 1 \le P_n(x) < \infty & x \le -1, & n = 0,2,4,\cdots \\ -\infty < P_n(x) \le -1 & x \le -1, & n = 1,3,5\cdots \end{cases}$$

The polynomial $P_n(x)$ has exactly n zeros and $(n-1)$ extrema; they all lie within the $-1 < x < 1$ zone.

Apart from the $n = 0$ case, $P_n(x)$ increases in magnitude without limit outside the $|x| < 1$ region, as $|x|$ increases, the rate of increase being greater the larger n is.

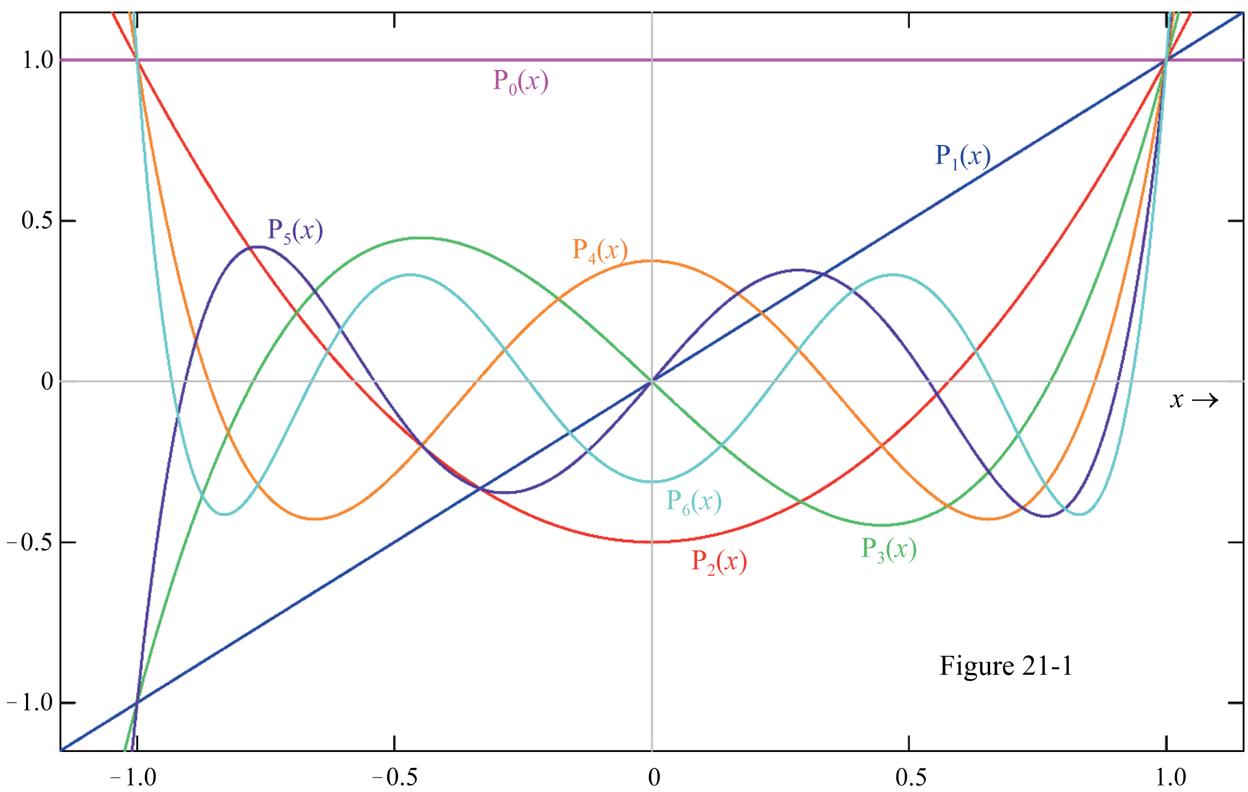


Figure 21-1

21:3 DEFINITIONS

Legendre polynomials may be defined by the generating function

21:3:1
$$\frac{1}{\sqrt{1-2xt+t^2}} = \left\{ \begin{matrix} \sum_{n=0}^{\infty} \mathrm{P}_n(x)t^n & -1<t<1 \\ \sum_{n=0}^{\infty} \mathrm{P}_n(x)t^{-n-1} & |t|>1 \end{matrix} \right\} -1 \le x \le 1$$

or by *Rodrigues's formula* (Benjamin Olinde Rodrigues, French mathematician, 1794 – 1851)

21:3:2
$$\mathrm{P}_n(x) = \frac{1}{2^n n!}\frac{\mathrm{d}^n}{\mathrm{d}x^n}\left(x^2-1\right)^n$$

One of many integral representations of Legendre polynomials is

21:3:3
$$\mathrm{P}_n(x) = \frac{1}{\pi}\int_0^{\pi}\left[x+\sqrt{x^2-1}\cos(t)\right]^n \mathrm{d}t \qquad x>1$$

known as *Laplace's representation*.

Because the Legendre polynomial is a hypergeometric function [Section 18:14]

21:3:4
$$\mathrm{P}_n(1-2x) = \sum_{j=0}^{\infty}\frac{(-n)_j(n+1)_j}{(1)_j(1)_j}x^j$$

it may be synthesized [Section 43:14] by the two-step process

21:3:5
$$\frac{1}{1-x} \xrightarrow[1]{-n} \left(1-x\right)^n \xrightarrow[1]{1+n} \mathrm{P}_n(1-2x)$$

from the basis function $1/(1-x)$.

The Legendre polynomial $\mathrm{f}(x) = \mathrm{P}_n(x)$ satisfies *Legendre's differential equation*

21:3:6
$$(1-x^2)\frac{\mathrm{d}^2\mathrm{f}}{\mathrm{d}x^2} - 2x\frac{\mathrm{d}\mathrm{f}}{\mathrm{d}x} + n(n+1)\mathrm{f} = 0$$

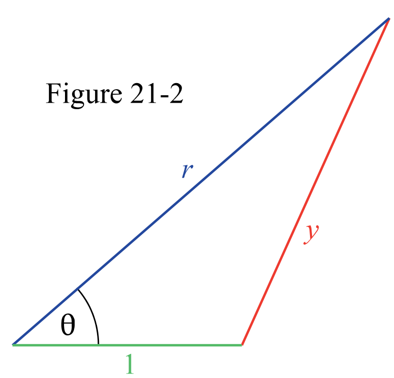
The most general solution of this equation is $\mathrm{f}(x) = w_1\mathrm{P}_n(x)+w_2\mathrm{Q}_n(x)$ where the w's are arbitrary weights and $\mathrm{Q}_n(x)$ is the function addressed in Section 21:13. Alternatively, the general solution of 21:3:6 may be written as the arbitrarily weighted sums of two series, namely those given by equation 21:6:1.

Figure 21-2 illustrates a geometric definition that explains how Legendre polynomials arise in certain applications. The triangle shown has two sides, one of unity length and one of length r, enclosing an angle θ. By the *law of cosines* [Section 34:14], the length y of the third side equals $\sqrt{1-2r\cos(\theta)+r^2}$. In some physical problems, it is necessary to expand $1/y$ as a power series in reciprocal powers of r. Definition 21:3:1 shows that the appropriate series is

21:3:7
$$\frac{1}{y} = \frac{\mathrm{P}_0\left(\cos(\theta)\right)}{r} + \frac{\mathrm{P}_1\left(\cos(\theta)\right)}{r^2} + \frac{\mathrm{P}_2\left(\cos(\theta)\right)}{r^3} + \cdots$$

if $r > 1$.

Figure 21-2



21:4 SPECIAL CASES

$P_0(x)$	$P_1(x)$	$P_2(x)$	$P_3(x)$	$P_4(x)$	$P_5(x)$	$P_6(x)$
1	x	$\frac{3}{2}x^2-\frac{1}{2}$	$\frac{5}{2}x^3-\frac{3}{2}x$	$\frac{35}{8}x^4-\frac{15}{4}x^2+\frac{3}{8}$	$\frac{63}{8}x^5-\frac{35}{4}x^3+\frac{15}{8}x$	$\frac{231}{16}x^6-\frac{315}{16}x^4+\frac{105}{16}x^2-\frac{5}{16}$

When an angular argument is adopted, it is often more convenient to use the cosines of multiple angles, rather than powers, as the expansion terms; for example

21:4:1 $$P_4\left(\cos(\theta)\right)=\left[9+20\cos(2\theta)+35\cos(4\theta)\right]/64$$

Gradshteyn and Ryzhik [Section 8.91] and Jeffrey [Section 18.2.4.2] list the corresponding expressions for other degrees.

21:5 INTRARELATIONSHIPS

Legendre polynomials are even or odd

21:5:1 $$P_n(-x)=(-)^{\left|n+\frac{1}{2}\right|-\frac{1}{2}}P_n(x)$$

according to the parity of n. The recursion formula

21:5:2 $$P_n(x)=\frac{2n-1}{n}xP_{n-1}(x)-\frac{n-1}{n}P_{n-2}(x) \qquad n=2,3,4,\cdots$$

relates three polynomials of consecutive degrees.

Section 21:4 lists expressions for $P_n(x)$ functions as finite series in powers of x. The converse is also possible: powers may be expressed as finite series of Legendre polynomials. For example

21:5:3 $$x^4=\tfrac{1}{5}P_0(x)+\tfrac{4}{7}P_2(x)+\tfrac{8}{35}P_4(x)$$

The general formula, which employs Pochhammer polynomials, is

21:5:4 $$x^n=\begin{cases}\displaystyle\sum_{j=0}^{n/2}\left[\frac{4j+1}{n+1}\right]\frac{\left(\frac{n+2}{2}-j\right)_j}{\left(\frac{n+3}{2}\right)_j}P_{2j}(x) & n=0,2,4,\cdots \\ \displaystyle\sum_{j=0}^{(n-1)/2}\left[\frac{4j+3}{n+2}\right]\frac{\left(\frac{n+1}{2}-j\right)_j}{\left(\frac{n+4}{2}\right)_j}P_{2j+1}(x) & n=1,3,5,\cdots\end{cases}$$

Not only powers, but a large number of functions $f(x)$ of x can be expressed as series of Legendre polynomials by exploiting the technique described in Section 21:15.

Among summation formulas for Legendre polynomials are

21:5:5 $$P_0(x)+3P_1(x)+\cdots+(2n-3)P_{n-2}(x)+(2n-1)P_{n-1}(x)=\frac{n[P_{n-1}(x)-P_n(x)]}{1-x}$$

and

21:5:6 $$P_0(x)-3P_1(x)+\cdots\pm(2n-3)P_{n-2}(x)\mp(2n-1)P_{n-1}(x)=\frac{\mp n[P_{n-1}(x)+P_n(x)]}{1+x}$$

21:6 EXPANSIONS

There are numerous power-series expansions of Legendre polynomials, all of which terminate; for example

21:6:1 $$P_n(x)=\begin{cases}\dfrac{(-)^{n/2}(n-1)!!}{n!!}\left[1-\dfrac{n(n+1)}{2!}x^2+\dfrac{(n-2)n(n+1)(n+3)}{4!}x^4-\cdots \qquad n=0,2,4,\cdots\right]\\[2ex] \dfrac{(-)^{(n-1)/2}n!!}{(n-1)!!}\left[x-\dfrac{(n-1)(n+2)}{3!}x^3+\dfrac{(n-3)(n-1)(n+2)(n+4)}{5!}x^5-\cdots \qquad n=1,3,5,\cdots\right]\end{cases}$$

There are many ways in which the Legendre polynomials may be written as Gauss hypergeometric functions [Chapter 60]; the simplest is known as *Murphy's formula*, which leads to a terminating power series in $1-x$:

21:6:2 $$P_n(x)=F\left(-n,1+n;1;\frac{1-x}{2}\right)=1-\frac{n(n+1)}{2}(1-x)+\frac{(n-1)n(n+1)(n+2)}{16}(1-x)^2-\cdots+\frac{(2n)!}{(n!)^2}\left(\frac{x-1}{2}\right)^n$$

More are listed by Gradshteyn and Ryzhik [Section 8.91], or may be found by specializing formulas from Chapter 59 to integer degree.

The infinite Fourier series

21:6:3 $$P_n(\cos(\theta))=\frac{4}{\pi}\frac{(2n)!!}{(2n+1)!!}\sum_{j=0}^{\infty}\frac{(2j-1)!!(n+1)_j}{(2j)!!(n+\frac{3}{2})_j}\sin((n+2j+1)\theta)$$

involves coefficients that are quotients of Pochhammer polynomials [Chapter 18] and double factorials [Section 2:13].

21:7 PARTICULAR VALUES

Legendre polynomials in the $n=2,6,10,\cdots$ family display a minimum at $x=0$, whereas a maximum is exhibited there if n is a multiple of 4. Away from $x=0$, we know of no general formulas for the zeros or extrema of $P_n(x)$.

	$P_n(-1)$	$P_n(0)$	$P_n(1)$
$n=0,2,4,\cdots$	1	$\dfrac{(-)^{n/2}(n-1)!!}{n!!}$	1
$n=1,3,5,\cdots$	-1	0	1

21:8 NUMERICAL VALUES

Equator's Legendre P polynomial routine (keyword **Ppoly**) uses recursion formula 21:5:2, initialized by $P_0(x)=1$ and $P_1(x)=x$.

21:9 LIMITS AND APPROXIMATIONS

The approximation [Lebedev]

21:9:1 $$P_n(\cos(\theta))\approx\sqrt{\frac{2}{n\pi\sin(\theta)}}\sin\left\{\left(n+\frac{1}{2}\right)\theta+\frac{\pi}{4}\right\}$$

holds for large n when $0<\theta<\pi$.

21:10 OPERATIONS OF THE CALCULUS

Differentiation and integration of a Legendre polynomial give

21:10:1
$$\frac{d}{dx}P_n(x) = \frac{n}{1-x^2}\left[P_{n-1}(x) - xP_n(x)\right]$$

and

21:10:2
$$\int_{-1}^{x} P_n(t)\,dt = \frac{P_{n+1}(x) - P_{n-1}(x)}{2n+1} \qquad n = 1,2,3,\cdots$$

Equation 21:10:1 is just one of a number of formulas relating the Legendre polynomials to its derivative. Others include

21:10:3
$$x\frac{d}{dx}P_n(x) - \frac{d}{dx}P_{n-1}(x) = nP_n(x) \qquad n = 1,2,3,\cdots$$

and

21:10:4
$$\frac{d}{dx}\{P_{n+1}(x) - P_{n-1}(x)\} = (2n+1)P_n(x) \qquad n = 1,2,3,\cdots$$

There are many useful definite integrals involving Legendre polynomials. These include

21:10:5
$$\int_{-1}^{1} P_n(t)\,dt = 0 \qquad n = 1,2,3,\cdots$$

21:10:6
$$\int_{-1}^{1} \frac{P_n(t)}{\sqrt{1-t}}\,dt = \frac{\sqrt{8}}{2n+1} \qquad n = 0,1,2,\cdots$$

21:10:7
$$\int_{-1}^{1} \frac{P_n(t)}{\sqrt{1-t^2}}\,dt = \begin{cases} \pi\left[(n-1)!!/n!!\right]^2 & n = 0,2,4,\cdots \\ 0 & n = 1,3,5,\cdots \end{cases}$$

and

21:10:8
$$\int_{0}^{1} t^{\mu}P_n(t)\,dt = \frac{(\mu-n+2)(\mu-n+4)(\mu-n+6)\cdots(\mu-\frac{1}{2}\pm\frac{1}{2})}{(\mu+n+1)(\mu+n-1)(\mu+n-3)\cdots(\mu+\frac{3}{2}\mp\frac{1}{2})} \qquad \mu > \pm\tfrac{1}{2}-\tfrac{3}{2}$$

where, in the final example, the upper/lower signs apply according as n is even/odd. Gradshteyn and Ryzhik devote several pages [Sections 7.22−7.25] to a listing of further integrals involving Legendre polynomials. The integral, between −1 and +1, of the product of a Legendre polynomial and some other function may often be evaluated by recourse to Rodrigues's definition 21:3:2, with help from parts integration [Section 0:10].

Legendre functions are orthogonal [Section 21:14] on the interval between −1 and +1 with a weight function of unity:

21:10:9
$$\int_{-1}^{1} P_n(t)P_m(t)\,dt = \begin{cases} 0 & m \ne n \\ \dfrac{2}{2n+1} & m = n \end{cases}$$

Laplace transforms involving the Legendre polynomial include

21:10:10
$$\int_{0}^{\infty} P_n(1+t)\exp(-st)\,dt = \mathfrak{L}\{P_n(1+t)\} = \sqrt{\frac{2}{\pi s}}\exp(s)K_{(2n+1)/2}(s)$$

21:10:11
$$\int_0^\infty \mathrm{P}_n(1-t)\exp(-st)\,\mathrm{d}t = \mathfrak{L}\{\mathrm{P}_n(1-t)\} = \sum_{j=0}^{n} \frac{(-n)_j(1+n)_j}{(1)_j}\left(\frac{1}{2s}\right)^j = \frac{1}{\pi s}\exp\left(\frac{1}{2s}\right)\mathrm{i}_n\left(\frac{1}{2s}\right)$$

and

21:10:12
$$\int_0^\infty (t+1)^n \mathrm{P}_n\left(\frac{t-1}{t+1}\right)\exp(-st)\,\mathrm{d}t = \mathfrak{L}\left\{(t+1)^n\mathrm{P}_n\left(\frac{t-1}{t+1}\right)\right\} = \frac{n!}{s^{n+1}}\mathrm{L}_n(s)$$

Respectively, functions from Chapter 51, Section 28:13, and Chapter 23 appear in these three transforms. The Hilbert transform [Section 7:10] of a Legendre polynomial

21:10:13
$$\frac{1}{\pi}\int_{-1}^{1}\mathrm{P}_n(t)\frac{\mathrm{d}t}{t-y} = \frac{-2}{\pi}\mathrm{Q}_n(y)$$

generates a Legendre function of the second kind [Section 21:13] of the same integer degree, a result known as *Neumann's formula*.

21:11 COMPLEX ARGUMENT

Although most applications of Legendre polynomials require a real argument, there is some interest in $\mathrm{P}_n(z)$ where z is a complex variable. One definition, the *Schläfli representation* (Ludwig Schläfli, Swiss theologian and mathematician, 1814 – 1895) employs the contour integral

21:11:1
$$\mathrm{P}_n(z) = \frac{1}{2^n}\oint \frac{(t^2-1)^n}{(t-z)^{n+1}}\frac{\mathrm{d}t}{2\pi i}$$

where the contour encloses the point z once.

Two formulas for Laplace inversion are

21:11:2
$$\int_{\alpha-i\infty}^{\alpha+i\infty}\frac{1}{s}\mathrm{P}_n\left(1-\frac{a}{s}\right)\frac{\exp(ts)}{2\pi i}\mathrm{d}s = \mathfrak{I}\left\{\frac{1}{s}\mathrm{P}_n\left(1-\frac{a}{s}\right)\right\} = \frac{1}{a}\sum_{j=0}^{n}\frac{(-n)_j(1+n)_j}{(1)_j(1)_j(1)_j}\left(\frac{at}{2}\right)^j$$

and

21:11:3
$$\int_{\alpha-i\infty}^{\alpha+i\infty}\frac{1}{s^{n+1}}\mathrm{P}_n\left(1-\frac{a}{s}\right)\frac{\exp(ts)}{2\pi i}\mathrm{d}s = \mathfrak{I}\left\{\frac{1}{s^{n+1}}\mathrm{P}_n\left(1-\frac{a}{s}\right)\right\} = \frac{t^n}{2^n n!a}\mathrm{L}_n\left(\frac{at}{2}\right)$$

21:12 GENERALIZATIONS

A modest generalization of the $\mathrm{P}_n(x)$ polynomial is to admit negative integer degrees by the degree-reflection formula

21:12:1
$$\mathrm{P}_{-n}(x) = \mathrm{P}_{n-1}(x)$$

which is consistent with 21:1:1. More radical is the generalization of the degree to any real number, thereby creating the Legendre function of the first kind, $\mathrm{P}_\nu(x)$, discussed in Chapter 59. A still wider generalization is to the associated Legendre functions of the first kind, $\mathrm{P}_\nu^{(\mu)}(x)$, which are the subject of Section 59:13.

21:13 COGNATE FUNCTIONS

The $Q_n(x)$ functions are mentioned in the context of equations 21:3:6 and 21:10:13. They are sometimes called "*Legendre polynomials of the second kind*" but this is an unfortunate name because they are not polynomial functions. They are the integer-degree instances of *Legendre functions of the second kind*, addressed in more detail in Chapter 59. Some special cases, applicable when $-1 < x < 1$, are tabulated

$Q_0(x)$	$Q_1(x)$	$Q_2(x)$	$Q_3(x)$
$\mathrm{artanh}(x)$	$x\,\mathrm{artanh}(x)-1$	$\left(\frac{3}{2}x^2-\frac{1}{2}\right)\mathrm{artanh}(x)-\frac{3}{2}x$	$\left(\frac{5}{2}x^3-\frac{3}{2}x\right)\mathrm{artanh}(x)-\frac{5}{2}x^2+\frac{2}{3}$

and the general formula is

21:13:1 $$Q_n(x) = P_n(x)\mathrm{artanh}(x) - 2\sum_{j=1,3,5}^{J}\frac{2n-2j+1}{j(2n-j+1)}P_{n-j}(x) \qquad J = 1+2\mathrm{Int}\left(\frac{n-1}{2}\right)$$

Here, artanh(x) is the inverse hyperbolic tangent function [Chapter 31], being equivalent to $\ln\left(\sqrt{(1+x)/(1-x)}\right)$. These definitions apply only for $|x| < 1$; outside this range replace artanh(x) by arcoth(x).

In distinction from $P_n(x)$, $Q_n(x)$ has a discontinuity of the $+\infty|+\infty$ variety at $x = 1$, another discontinuity at $x = -1$, and, as Figure 21-3 suggests, an approach towards zero as $x \to \pm\infty$. Moreover, in contrast to $P_n(x)$, $Q_n(x)$ is an even function when n is odd, and vice versa. Nonetheless, the two functions do have much in common: equations 21:5:2 and 21:10:1−4 retain their validity when Q replaces P. Each satisfies Legendre's differential equation 21:3:7.

$Q_n(z)$ is multivalued in the complex plane, unless suitably cut. Usually a branch cut is applied along the real axis between −1 and +1. The average of the values on either side of the cut is assigned to $Q_n(x)$, while the difference of these values is $-\pi i P_n(x)$. See Erdélyi et al [*Higher Transcendental Functions*, Volume 2, Page 181].

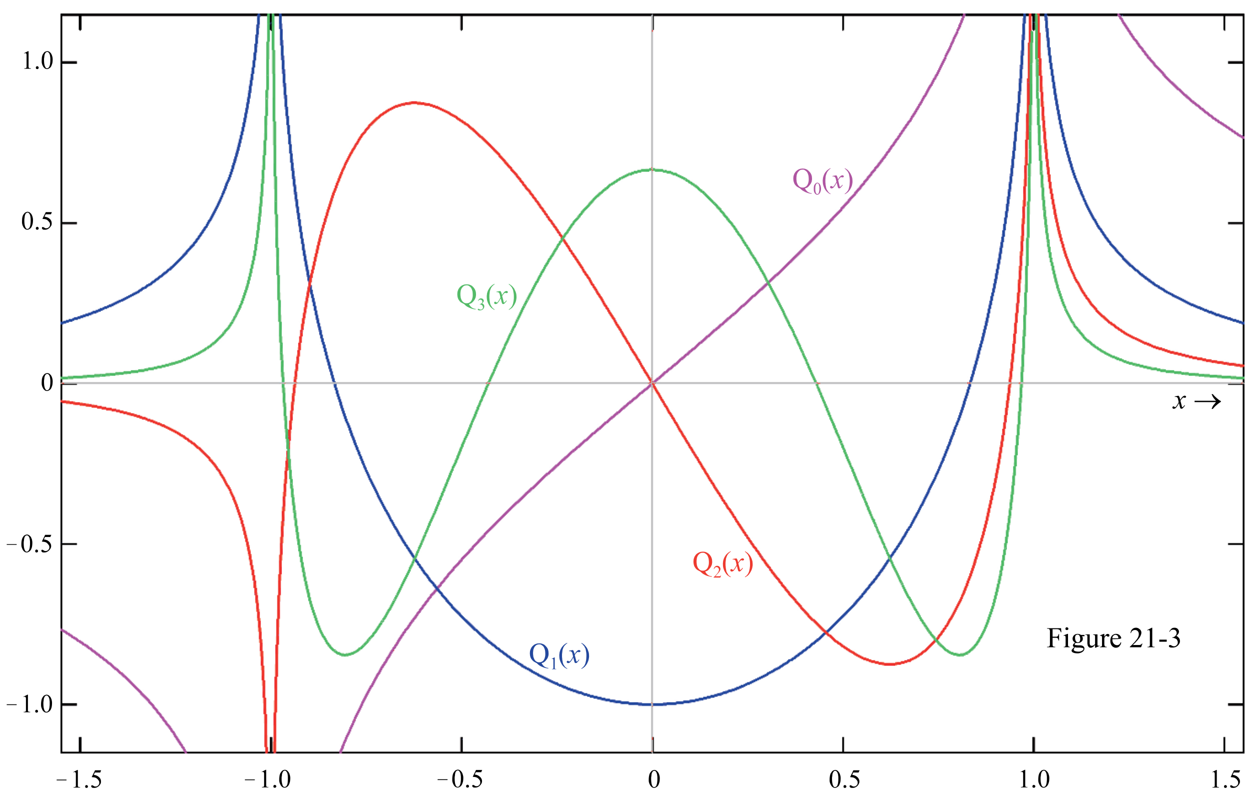


Figure 21-3

21:14 RELATED TOPIC: orthogonality

The integral, between distinct limits, of the square f × f of a real function f is never zero (unless the function is the zero function); in fact it must be positive. The definite integral of the product of two functions f × g is rarely zero

21:14:1
$$\int_{x_0}^{x_1} f(t)g(t)\,dt = 0 \qquad f \neq g$$

but this unusual property is possessed by any two Legendre polynomials, $P_n(x)$ and $P_m(x)$, if $x_0 = -1$ and $x_1 = 1$. Specifically, to repeat equation 21:10:10

21:14:2
$$\int_{-1}^{1} P_n(t)P_m(t)\,dt = \begin{Bmatrix} 0 & m \neq n \\ \dfrac{2}{2n+1} & m = n \end{Bmatrix} \quad m,n = 0,1,2,\cdots$$

The two Legendre polynomials are said to be mutually "orthogonal on the interval x_0 and x_1".

If a carefully chosen third function, called a *weight function* w(t) is introduced into the integrand, then the integral is more likely to vanish. If it does, that is if

21:14:3
$$\int_{x_0}^{x_1} w(t)f(t)g(t)\,dt = 0 \qquad f \neq g$$

then one says that "f and g are orthogonal on the interval x_0 to x_1 with respect to the weight function w". The weight function is required to be nonnegative throughout the x_0 to x_1 interval. Legendre polynomials are orthogonal on the interval −1 to +1 with a weight function of unity.

Many polynomial functions, including those in Chapters 21 through 24 of this *Atlas*, constitute *orthogonal families*. The accompanying table shows some of the orthogonality properties of the Legendre, Chebyshev, Jacobi, Gegenbauer, Laguerre, and Hermite polynomials.

$\Psi_n(t)$	x_0 to x_1	w(t)	Ω_n^2
$P_n(t)$	−1 to 1	1	$2/(2n+1)$
$T_n(t)$	−1 to 1	$1/\sqrt{1-t^2}$	$\pi/2$ (or π if $n=0$)
$U_n(t)$	−1 to 1	$\sqrt{1-t^2}$	$\pi/2$
$P_n(2t-1)$	0 to 1	$1/\sqrt{1-t^2}$	$1/(2n+1)$
$T_n(2t-1)$	0 to 1	$1/\sqrt{1-t^2}$	$\pi/2$ (or π if $n=0$)
$U_n(2t-1)$	0 to 1	$\sqrt{t-t^2}$	$\pi/8$
$P_n^{(\nu,\mu)}(t)$	−1 to 1	$\dfrac{(1-t)^\nu}{(1+t)^\mu}$	$\dfrac{2^{\nu+\mu+1}\Gamma(n+\nu+1)\Gamma(n+\mu+1)}{(2n+\nu+\mu+1)n!\Gamma(n+\nu+\mu+1)}$
$C_n^{(\lambda)}(t)$	−1 to 1	$(1-t^2)^{(2\lambda-1)/2}$	$\dfrac{2^{1-2\lambda}\pi\Gamma(n+2\lambda)}{(n+\lambda)n!\Gamma^2(\lambda)}$
$L_n(x)$	0 to ∞	$\exp(-t)$	1
$H_n(x)$	−∞ to ∞	$\exp(-t^2)$	$2^n n!\sqrt{\pi}$

A general definition is that the family of functions $\Psi_0(x)$, $\Psi_1(x)$, $\Psi_2(x)$, ··· are orthogonal when, for all choices of nonnegative integers n and m

21:14:4 $$\int_{x_0}^{x_1} w(t)\Psi_n(t)\Psi_m(t)\,dt = \begin{cases} 0 & m \neq n \\ \Omega_n^2 \neq 0 & m = n \end{cases}$$

where x_0, x_1, and w are specified. An alternative way of formulating this equation is

21:14:5 $$\int_{x_0}^{x_1} w(t)\left[\frac{\Psi_n(t)}{\Omega_n}\right]\left[\frac{\Psi_m(t)}{\Omega_m}\right] dt = \delta_{n,m}$$

where $\delta_{n,m}$ is the Kronecker delta function [Section 9:13]. The Ω terms are normalizing factors and $\Psi_n(x)/\Omega_n$ is the *normalized* or *orthonormal* version of the nth orthogonal polynomial. The normalized version of the Legendre polynomial is $\sqrt{n+\frac{1}{2}}\,P_n(x)$.

Orthogonality is not restricted to polynomials. How the sine and cosine functions group into orthogonal families is explained in Section 32:10.

From a weighted sum of orthogonal functions one may extract the coefficient of one particular, say the mth, summand by the integration procedure

21:14:6 $$\int_{x_0}^{x_1}\left[\sum_n a_n\Psi_n(t)\right]\Psi_m(t)w(t)\,dt = a_m\Omega_m^2$$

This is one of the prime benefits conferred by orthogonality.

21:15 RELATED TOPIC: expansions in Legendre functions

The orthogonality property of the orthogonal polynomials $\Psi_0(x)$, $\Psi_1(x)$, $\Psi_2(x)$, $\cdots$ permits their use as a *basis set* for the expansion

21:15:1 $$f(x) = c_0\Psi_0(x) + c_1\Psi_1(x) + c_2\Psi_2(x) + \cdots$$

for a wide range of f(x) functions. Such an expansion, which is called a *orthogonal series* or a *generalized Fourier expansion* is an alternative to, and in some circumstances an improvement on, the most usual basis set, the powers x^0, x^1, x^2, $\cdots$. The c's are the coefficients of the expansion. Multiplying equation 21:15:1 by $w(x)\Psi_n(x)$ and integrating shows that the coefficients may be found from the integral

21:15:2 $$c_n = \frac{1}{\Omega_n^2}\int_{x_0}^{x_1} f(t)w(t)\Psi_n(t)\,dt$$

Any orthogonal family can be used in this context, but the Legendre polynomials are often the most convenient because their weight function is unity. For this basis set,

21:15:3 $$c_n = \frac{2n+1}{2}\int_{-1}^{1} f(t)P_n(t)\,dt$$

Two examples of series constructed in this way are

21:15:4 $$\sqrt{a^2-x^2} = \frac{\pi a}{2}\left[\frac{1}{2} - \sum_{j=2,4}^{\infty}(2j+1)\frac{(j-3)!!(j-1)!!}{j!!(j+1)!!}P_j\left(\frac{x}{a}\right)\right]$$

and

21:15:5 $$\frac{1}{\sqrt{a^2-x^2}} = \frac{\pi}{2a}\sum_{j=0,2}^{\infty}(2j+1)\left[\frac{(j-1)!!}{j!!}\right]^2 P_j\left(\frac{x}{a}\right)$$

while a third is provided by equation 35:6:5. In these equations, the coefficients are written as double factorials.

CHAPTER 22

THE CHEBYSHEV POLYNOMIALS $\mathrm{T}_n(x)$ AND $\mathrm{U}_n(x)$

Named for the Russian mathematician Pafnuty Lvovich Chebyshev (1821−1894), these two kinds of polynomial function are interrelated in the many ways detailed in Section 22:5 and by several of the integrals in Section 22:10. $\mathrm{T}_n(x)$ is extensively used in fitting procedures of various kinds. Gegenbauer and Jacobi polynomials are also briefly addressed in this chapter, as are discrete Chebyshev polynomials.

22:1 NOTATION

Alternative transliterations from the Cyrillic alphabet lead to spelling variants ranging from Chebyshev to Tschebischeff.

Unfortunately, mathematicians differ in their definitions of Chebyshev polynomials, and there is not even unanimity on which of the two kinds constitutes the "first" family and which the "second". The notations $\mathrm{T}_n(x)$ and $\mathrm{U}_n(x)$ may be encountered with meanings equivalent to the $\mathrm{T}_n(x)/2^{n-1}$ and $n\mathrm{U}_n(x)$ of this *Atlas*. Moreover, the symbol $\mathrm{U}_n(x)$ and the name "Chebyshev polynomial of the second kind" is commonly applied to a set of functions defined by $\arcsin(x)$ when $n = 0$, and by our $\sqrt{1-x^2}\,\mathrm{U}_{n-1}(x)$ when $n = 1,2,3,\cdots$, despite these not being polynomials at all.

Several supplementary notations may be encountered, but none of these is employed in the *Atlas*. Thus $\mathrm{T}_n^*(x)$ and $\mathrm{U}_n^*(x)$ are often used to symbolize the *shifted Chebyshev polynomials* $\mathrm{T}_n(2x-1)$ and $\mathrm{U}_n(2x-1)$, cited in the table in Section 21:14. The symbols $\mathrm{C}_n(x)$ and $\mathrm{S}_n(x)$ have been used for $2\mathrm{T}_n(x/2)$ and $\mathrm{U}_n(x/2)$, respectively. Under the name *Chebyshev functions*, $\overline{\mathrm{T}}_\nu(x)$ and $\overline{\mathrm{U}}_\nu(x)$ symbolize functions resembling $\mathrm{T}_n(x)$ and $\mathrm{U}_n(x)$ but in which the degree is not restricted to integer values.

Here, we shall name $\mathrm{T}_n(x)$ the first, and $\mathrm{U}_n(x)$ the second, Chebyshev polynomial family. The degree n takes the nonnegative integer values $0,1,2,\cdots$, though $\mathrm{U}_0(x)$ is sometimes excluded from the second family when orthogonality is an issue. The argument x is unrestricted, but interest is concentrated in the $-1 \le x \le 1$ range.

K.B. Oldham et al., *An Atlas of Functions, Second Edition*,
DOI 10.1007/978-0-387-48807-3_23, © Springer Science+Business Media, LLC 2009

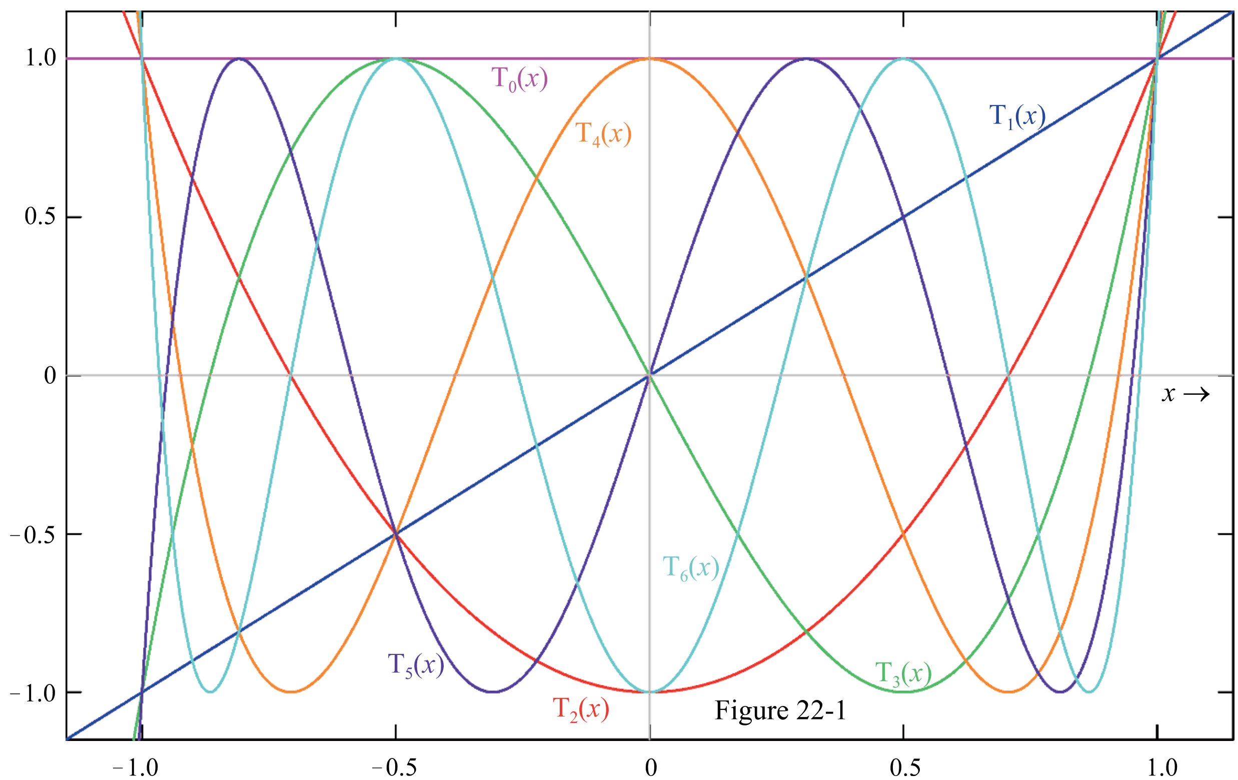


Figure 22-1

22:2 BEHAVIOR

Figures 22-1 and 22-2 portray $T_n(x)$ and $U_n(x)$ respectively. Each kind of Chebyshev polynomial has precisely n zeros, Int(n/2) minima and Int$\{(n-1)/2\}$ maxima, all these features lying within $-1 < x < 1$. The locations of the zeros are given in Section 22:7. All the maxima of $T_n(x)$ take the value +1, whereas $T_n(x)$ equals -1 at all the minima; no comparable rules apply to $U_n(x)$. For $-1 \le x \le 1$ range, the ranges of the two polynomials are

22:2:1 $$-1 \le T_n(x) \le 1$$

22:2:2 $$-n-1 \le U_n(x) \le n+1 \qquad \left\{ \begin{array}{l} -1 \le x \le 1 \\ n = 1,2,3,\cdots \end{array} \right.$$

22:3 DEFINITIONS

Some, but not all, of these definitions apply outside the $-1 \le x \le 1$ range of interest.

Purely algebraic definitions of the polynomial families are

22:3:1 $$T_n(x) = \frac{1}{2}\left[\left(x + i\sqrt{1-x^2}\right)^n + \left(x - i\sqrt{1-x^2}\right)^n\right]$$

22:3:2 $$U_n(x) = \frac{1}{2i\sqrt{1-x^2}}\left[\left(x + i\sqrt{1-x^2}\right)^{n+1} - \left(x - i\sqrt{1-x^2}\right)^{n+1}\right]$$

but the definitions most usually encountered are trigonometrically based:

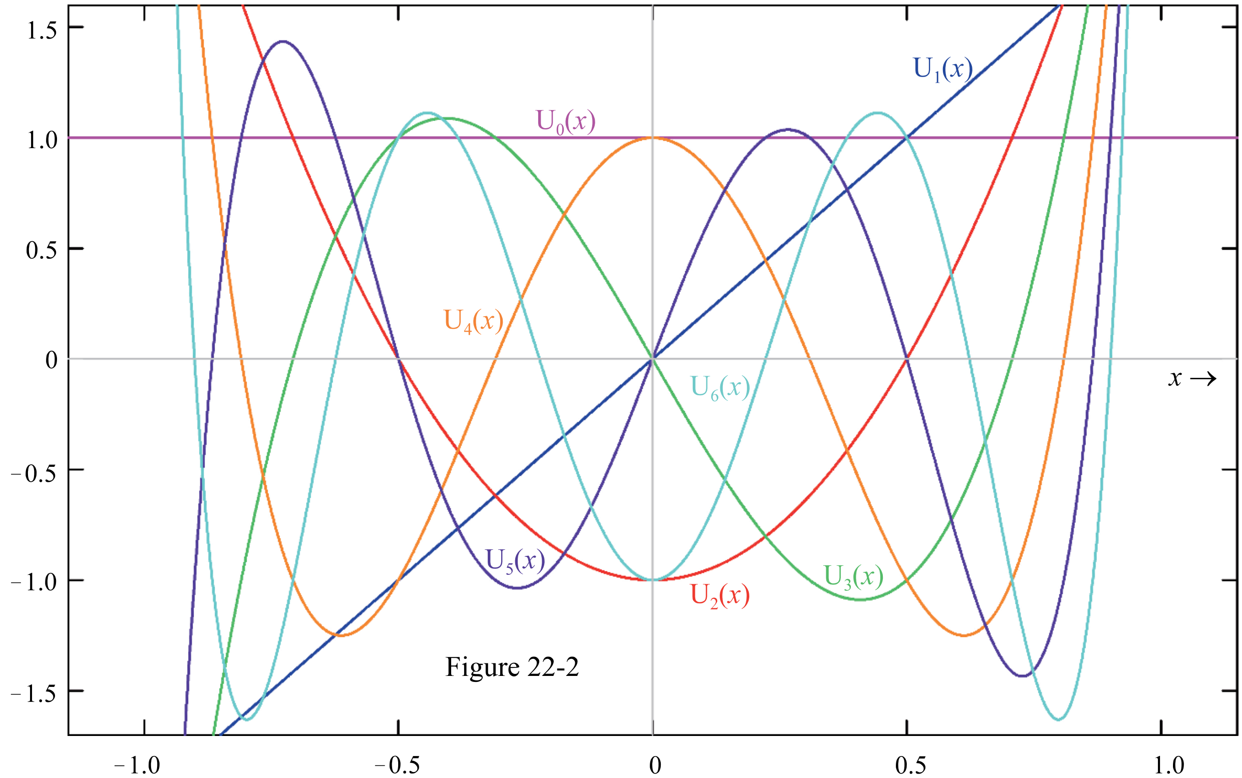


Figure 22-2

22:3:3 $$T_n(x) = \cos(n\theta)$$

22:3:4 $$U_n(x) = \csc(\theta)\sin\{(n+1)\theta\}$$

$$\theta = \arccos(x) \qquad -1 \le x \le 1$$

The Chebyshev polynomials may be written as Gauss hypergeometric functions [Chapter 60], or equivalently as the straightforward hypergeometric functions

22:3:5 $$T_n(1-2x) = \sum_{j=0}^{n} \frac{(-n)_j (n)_j}{(\tfrac{1}{2})_j (1)_j} x^j$$

and

22:3:6 $$U_n(1-2x) = (n+1)\sum_{j=0}^{n} \frac{(-n)_j (n+2)_j}{(1)_j (\tfrac{3}{2})_j} x^j$$

They may be synthesized by routes analogous to that in 21:3:5.

Chebyshev polynomials may be defined by the generating functions

22:3:7 $$\frac{1-tx}{1-2tx+t^2} = \sum_{n=0}^{\infty} T_n(x) t^n$$

22:3:8 $$\frac{1}{1-2tx+t^2} = \sum_{n=0}^{\infty} U_n(x) t^n$$

or by the *Rodrigues's formulas*

22:3:9 $$T_n(x) = \frac{(-)^n \sqrt{1-x^2}}{(2n-1)!!} \frac{d^n}{dx^n} \left(1-x^2\right)^{(2n-1)/2}$$

22:3:10 $$U_n(x) = \frac{(-)^n(n+1)}{(2n-1)!!\sqrt{1-x^2}}\frac{d^n}{dx^n}\left(1-x^2\right)^{(2n+1)/2}$$

For $n = 1,2,3,\cdots$, a general solution to *Chebyshev's differential equation*

22:3:11 $$(1-x^2)\frac{d^2f}{dx^2} - x\frac{df}{dx} + n^2 f = 0$$

is

22:3:12 $$f(x) = w_1 T_n(x) + w_2\sqrt{1-x^2}\,U_{n-1}(x)$$

where the w's are arbitrary. For $n = 0$, the solution is $f(x) = w_1 + w_2 \arcsin(x)$.

22:4 SPECIAL CASES

$T_0(x)$	$T_1(x)$	$T_2(x)$	$T_3(x)$	$T_4(x)$	$T_5(x)$	$T_6(x)$	$T_7(x)$
1	x	$2x^2-1$	$4x^3-3x$	$8x^4-8x^2+1$	$16x^5-20x^3+5x$	$32x^6-48x^4+18x^2-1$	$64x^7-112x^5+56x^3-7x$

$U_0(x)$	$U_1(x)$	$U_2(x)$	$U_3(x)$	$U_4(x)$	$U_5(x)$	$U_6(x)$	$U_7(x)$
1	$2x$	$4x^2-1$	$8x^3-4x$	$16x^4-12x^2+1$	$32x^5-32x^3+6x$	$64x^6-80x^4+24x^2-1$	$128x^7-192x^5+80x^3-8x$

22:5 INTRARELATIONSHIPS

Chebyshev polynomials are even or odd

22:5:1 $$f_n(-x) = (-)^n f_n(x) \qquad f = T \text{ or } U$$

according to the parity of their degree. The recursion formula

22:5:2 $$f_n(x) = 2x f_{n-1}(x) - f_{n-2}(x) \qquad n = 2,3,4,\cdots$$

also applies equally to both kinds of Chebyshev polynomial.

The formulas

22:5:3 $$T_n(x) = U_n(x) - xU_{n-1}(x) \qquad n = 1,2,3,\cdots$$

and

22:5:4 $$U_n(x) = \frac{T_n(x) - xT_{n+1}(x)}{1-x^2} \qquad n = 0,1,2,\cdots$$

permit one kind of Chebyshev polynomial to be expressed in terms of the other. There are also formulas expressing the product of two Chebyshev polynomials as the sum of other Chebyshev polynomials:

22:5:5 $$T_n(x)T_m(x) = \frac{T_{n+m}(x) + T_{n-m}(x)}{2} \qquad n \ge m$$

22:5:6 $$U_n(x)U_m(x) = \frac{T_{n-m}(x) - T_{n+m+2}(x)}{2(1-x^2)} \qquad n \ge m$$

22:5:7
$$T_n(x)U_m(x) = \begin{cases} \frac{1}{2}U_{n+m}(x) + \frac{1}{2}U_{n-m}(x) & n \le m \\ \frac{1}{2}U_{n+m}(x) & n = m+1 \\ \frac{1}{2}U_{n+m}(x) - \frac{1}{2}U_{n-m-2}(x) & n \ge m+2 \end{cases}$$

Series of Chebyshev polynomials of the first kind have the sums

22:5:8
$$T_0(x) + T_2(x) + T_4(x) + \cdots + T_n(x) = \frac{1}{2} + \frac{1}{2}U_n(x) \qquad n = 0,2,4,\cdots$$

and

22:5:9
$$T_1(x) + T_3(x) + T_5(x) + \cdots + T_n(x) = \frac{1}{2}U_n(x) \qquad n = 1,3,5,\cdots$$

Clearly, these two expressions may be combined additively or subtractively to produce other useful sums. The corresponding series for Chebyshev polynomials of the second kind are

22:5:10
$$U_0(x) + U_2(x) + U_4(x) + \cdots + U_n(x) = \frac{1 - T_{n+2}(x)}{2(1-x^2)} \qquad n = 0,2,4,\cdots$$

and

22:5:11
$$U_1(x) + U_3(x) + U_5(x) + \cdots + U_n(x) = \frac{x - T_{n+2}(x)}{2(1-x^2)} \qquad n = 1,3,5,\cdots$$

Positive integer powers of x can be expressed as finite series of Chebyshev polynomials of either kind. For example

22:5:12
$$x^3 = \frac{3}{4}T_1(x) + \frac{1}{4}T_3(x) = \frac{1}{4}U_1(x) + \frac{1}{8}U_3(x)$$

Restricting attention to the polynomials of the first kind, one has the general expression

22:5:13
$$x^n = \sum_{j=0}^{n} \gamma_j^{(n)} T_j(x) = \gamma_n^{(n)} T_n(x) + \gamma_{n-2}^{(n)} T_{n-2}(x) + \gamma_{n-4}^{(n)} T_{n-4}(x) + \cdots + \begin{cases} \gamma_0^{(n)} T_0(x) & n = 0,2,4,\cdots \\ \gamma_1^{(n)} T_1(x) & n = 1,3,5,\cdots \end{cases}$$

Note that $\gamma_j^{(n)} = 0$ if j and n have unlike parities, or if $j > n$. Values of these coefficients are provided by *Equator*'s Chebyshev gamma coefficient routine (keyword **Chebygamma**) which uses the following recursive algorithm, executed sequentially for $n = 0$, $n = 1$, $n = 2$, $n = 3$, $n = 4$, ⋯.

22:5:14
$$\begin{cases} n = 0,1 & \gamma_n^{(n)} = 1 \\ n = 2,4,6,\cdots & \begin{cases} \gamma_0^{(n)} = \frac{1}{2}\gamma_1^{(n-1)} \\ \gamma_j^{(n)} = \frac{1}{2}\gamma_{j-1}^{(n-1)} + \frac{1}{2}\gamma_{j+1}^{(n-1)} & \text{for } j = 2,4,6,\cdots,(n-2) \\ \gamma_n^{(n)} = \frac{1}{2}\gamma_{n-1}^{(n-1)} \end{cases} \\ n = 3,5,7,\cdots & \begin{cases} \gamma_1^{(n)} = \gamma_0^{(n-1)} + \frac{1}{2}\gamma_2^{(n-1)} \\ \gamma_j^{(n)} = \frac{1}{2}\gamma_{j-1}^{(n-1)} + \frac{1}{2}\gamma_{j+1}^{(n-1)} & \text{for } j = 3,5,7,\cdots,(n-2) \\ \gamma_n^{(n)} = \frac{1}{2}\gamma_{n-1}^{(n-1)} \end{cases} \end{cases}$$

Extensive tables of these, and similar, coefficients can be found in Abramowitz and Stegun [Chapter 22]. Here we list only coefficients of $T_0(x)$, $T_1(x)$, $T_2(x)$ and $T_3(x)$ in the expansions of x^0, x^1, x^2, ⋯, x^{13}.

	$n=0$	$n=1$	$n=2$	$n=3$	$n=4$	$n=5$	$n=6$	$n=7$	$n=8$	$n=9$	$n=10$	$n=11$	$n=12$	$n=13$
$\gamma_0^{(n)}$	1	0	$1/2$	0	$3/8$	0	$5/16$	0	$35/128$	0	$63/256$	0	$231/1024$	0
$\gamma_1^{(n)}$	0	1	0	$3/4$	0	$5/8$	0	$35/64$	0	$63/128$	0	$231/512$	0	$429/1024$
$\gamma_2^{(n)}$	0	0	$1/2$	0	$1/2$	0	$15/32$	0	$7/16$	0	$105/256$	0	$99/256$	0
$\gamma_3^{(n)}$	0	0	0	$1/4$	0	$5/16$	0	$21/64$	0	$21/64$	0	$165/512$	0	$1287/4096$

22:6 EXPANSIONS

Explicit power-series expressions for the two Chebyshev polynomials are

22:6:1 $$T_n(x) = \frac{n}{2}\sum_{j=0}^{\mathrm{Int}(n/2)} \frac{(-)^j}{n-j}\binom{n-j}{j}(2x)^{n-2j} \qquad n=1,2,3,\cdots$$

and

22:6:2 $$U_n(x) = \sum_{j=0}^{\mathrm{Int}(n/2)} (-)^j \binom{n-j}{j}(2x)^{n-2j} \qquad n=0,1,2,\cdots$$

but these may also be written in a binomial-expansion-like format as

22:6:3 $$T_n(x) = \binom{n}{0}x^n - \binom{n}{2}x^{n-2}(1-x^2) + \binom{n}{4}x^{n-4}(1-x^2)^2 - \cdots$$

and

22:6:4 $$U_{n-1}(x) = \binom{n}{1}x^{n-1} - \binom{n}{2}x^{n-3}(1-x^2) + \binom{n}{3}x^{n-5}(1-x^2)^2 - \cdots$$

Furthermore, series expansions may be developed from the hypergeometric representations 22:3:5 and 22:3:6.

Equation 22:6:3 may be redrafted as

22:6:5 $$T_n(x) = \sum_{k=n,n-2,n-4}^{0 \text{ or } 1} \tau_k^{(n)} x^k$$

Early values of the coefficients $\tau_k^{(n)}$ are

	$n=0$	$n=1$	$n=2$	$n=3$	$n=4$	$n=5$	$n=6$	$n=7$	$n=8$	$n=9$	$n=10$	$n=11$	$n=12$	$n=13$
$\tau_0^{(n)}$	1	0	−1	0	1	0	−1	0	1	0	−1	0	1	0
$\tau_1^{(n)}$	0	1	0	−3	0	5	0	−7	0	9	0	−11	0	13
$\tau_2^{(n)}$	0	0	2	0	−8	0	18	0	−32	0	50	0	−72	0
$\tau_3^{(n)}$	0	0	0	4	0	−20	0	56	0	−120	0	220	0	−364

Note that if n and k are of unlike parity, or if $k > n$, the $\tau_k^{(n)}$ coefficient is zero. For computational purposes, nonzero values of these coefficients in the power-series expansion of the first kind of Chebyshev polynomial are most easily found from the formula

22:6:6
$$\tau_k^{(n)} = \frac{(-)^{(n-k)/2}2^{k-1}n\left(\tfrac{1}{2}n+\tfrac{1}{2}k-1\right)!}{\left(\tfrac{1}{2}n-\tfrac{1}{2}k\right)!k!} = \frac{(-)^{(n-k)/2}2^{k}n}{n+k}\binom{\tfrac{1}{2}n+\tfrac{1}{2}k}{k}$$

except that $\tau_0^{(0)} = 1$. This is the formula used by *Equator*'s Chebyshev tau coefficient routine (keyword **Chebytau**) to compute the coefficients of x^k in the power-series expansion of $T_n(x)$.

22:7 PARTICULAR VALUES

	$T_n(-1)$	$T_n(0)$	$T_n(1)$
$n = 0,2,4,\cdots$	1	$(-1)^{n/2}$	1
$n = 1,3,5,\cdots$	-1	0	1

	$U_n(-1)$	$U_n(0)$	$U_n(1)$
$n = 0,2,4,\cdots$	$n+1$	$(-1)^{n/2}$	$n+1$
$n = 1,3,5,\cdots$	$-n-1$	0	$n+1$

The zeros of the Chebyshev polynomials occur at arguments given by

22:7:1
$$r_j = \cos\left\{\frac{(2j-1)\pi}{2n}\right\} \qquad j = 1,2,3,\cdots,n \qquad \text{where} \qquad T_n(r_j) = 0$$

and

22:7:2
$$r_j = \cos\left\{\frac{j\pi}{n+1}\right\} \qquad j = 1,2,3,\cdots,n \qquad \text{where} \qquad U_n(r_j) = 0$$

22:8 NUMERICAL VALUES

Equator's Chebyshev polynomial of the first kind and Chebyshev polynomial of the second kind routines (keywords **Tpoly** and **Upoly**) are both based on recursion relation 22:5:2. The sole difference between the routines is that the T routine is initialized by $f_0 = 1$, $f_1 = x$, whereas the U routine uses $f_0 = 1$, $f_1 = 2x$.

22:9 LIMITS AND APPROXIMATIONS

In the limits as $x \to \pm\infty$

22:9:1
$$T_n(x) \to 2^{n-1}x^n \qquad n = 1,2,3,\cdots$$

and

22:9:2
$$U_n(x) \to 2^n x^n \qquad n = 0,1,2,\cdots$$

22:10 OPERATIONS OF THE CALCULUS

Luke [pages 298−302] gives a plethora of information respecting differentiation and indefinite integration of the Chebyshev polynomials. Some of the simplest results are:

22:10:1
$$\frac{\mathrm{d}}{\mathrm{d}x}\mathrm{T}_n(x) = n\mathrm{U}_{n-1}(x) = \frac{n\left[\mathrm{T}_{n-1}(x) - x\mathrm{T}_n(x)\right]}{1-x^2} \qquad n=1,2,3,\cdots$$

22:10:2
$$\frac{\mathrm{d}}{\mathrm{d}x}\mathrm{U}_n(x) = \frac{nx\,\mathrm{U}_n(x) - (n+1)\mathrm{T}_{n+1}(x)}{1-x^2} \qquad n=1,2,3,\cdots$$

22:10:3
$$\int_0^x \mathrm{T}_n(t)\,\mathrm{d}t = \frac{n}{n^2-1}\left[\mathrm{T}_{n+1}(x) + \sin\left(\frac{n\pi}{2}\right)\right] - \frac{x}{n-1}\mathrm{T}_n(x) \qquad n=2,3,4,\cdots$$

22:10:4
$$\int_0^x \mathrm{U}_n(t)\,\mathrm{d}t = \frac{\mathrm{T}_{n+1}(x) + \sin(n\pi/2)}{n+1} \qquad n=0,1,2,\cdots$$

22:10:5
$$\int_x^1 \mathrm{U}_n(t)\,\mathrm{d}t = \frac{1-\mathrm{T}_{n+1}(x)}{n+1} \qquad n=0,1,2,\cdots$$

The definite integrals

22:10:6
$$\int_{-1}^1 \frac{\mathrm{T}_n(t)\mathrm{T}_m(t)}{\sqrt{1-t^2}}\,\mathrm{d}t = \begin{cases} 0 & m \neq n \\ \pi & m=n=0 \\ \pi/2 & m=n=1,2,3,\cdots \end{cases}$$

and

22:10:7
$$\int_{-1}^1 \sqrt{1-t^2}\,\mathrm{U}_n(t)\mathrm{U}_m(t)\,\mathrm{d}t = \begin{cases} 0 & m \neq n \\ \pi/2 & m=n=0,1,2,\cdots \end{cases}$$

establish the orthogonality of the Chebyshev polynomials [Section 21:14]. Other definite integrals include

22:10:8
$$\int_{-1}^1 \mathrm{T}_n^2(t)\,\mathrm{d}t = \frac{4n^2-2}{4n^2-1} \qquad n=0,1,2,\cdots$$

22:10:9
$$\int_0^1 \frac{\cos(bt)\mathrm{T}_n(t)}{\sqrt{1-t^2}}\,\mathrm{d}t = \frac{(-)^{n/2}\pi}{2}\mathrm{J}_n(b) \qquad b>0 \qquad n=0,2,4,\cdots$$

22:10:10
$$\int_0^1 \frac{\sin(bt)\mathrm{T}_n(t)}{\sqrt{1-t^2}}\,\mathrm{d}t = \frac{(-)^{(n-1)/2}\pi}{2}\mathrm{J}_n(b) \qquad b>0 \qquad n=1,3,5,\cdots$$

and others are listed by Gradshteyn and Ryzhik [Sections 7.34−7.36]. In these integrals J_n denotes the Bessel function of order n [Chapter 51].

Examples of Laplace transforms of Chebyshev polynomials are

22:10:11
$$\int_0^\infty \frac{\mathrm{T}_n(1-2t)}{\sqrt{t}}\exp(-st)\,\mathrm{d}t = \mathfrak{L}\left\{\frac{\mathrm{T}_n(1-2t)}{\sqrt{t}}\right\} = \sqrt{\frac{\pi}{s}}\sum_{j=0}^\infty \frac{(-n)_j(n)_j}{(1)_j}\left(\frac{1}{s}\right)^j$$

and

22:10:12
$$\int_0^\infty \sqrt{t}\,\mathrm{U}_n(1-2t)\exp(-st)\,\mathrm{d}t = \mathfrak{L}\left\{\sqrt{t}\,\mathrm{U}_n(1-2t)\right\} = (-)^{n/2}\frac{n}{2}\sqrt{\frac{\pi}{s^3}}\sum_{j=0}^\infty \frac{(1-n)_j(n+1)_j}{(1)_j}\left(\frac{1}{s}\right)^j$$

Though written as $L = K-1 = 1$ hypergeometric functions, the transforms are themselves polynomials (they are sometimes called *Rainville polynomials*). Each kind of Chebyshev polynomial, multiplied by its weight function, converts, on Hilbert transformation [Section 7:10], to the other kind:

22:10:13 $$\frac{1}{\pi}\int_{-1}^{1}\frac{T_n(t)}{\sqrt{1-t^2}}\frac{dt}{t-y}=U_{n-1}(y) \qquad -1<y<1$$

22:10:14 $$\frac{1}{\pi}\int_{-1}^{1}U_n(t)\sqrt{1-t^2}\,\frac{dt}{t-y}=T_{n+1}(y) \qquad -1<y<1$$

These last two integrals must be interpreted as their Cauchy limits [equation 7:10:5].

22:11 COMPLEX ARGUMENT

Chebyshev polynomials of complex argument are sometimes encountered. The real and imaginary parts of early $T_n(x+iy)$ polynomials are

	$T_0(x+iy)$	$T_1(x+iy)$	$T_2(x+iy)$	$T_3(x+iy)$	$T_4(x+iy)$
Re	1	x	$2x^2-2y^2-1$	$4x^3-12xy^2-3x$	$8x^4-48x^2y^2+8y^4-8x^2+8y^2+1$
Im	0	y	$4xy$	$12x^2y-4y^3-3y$	$32x^3y-32xy^3-16xy$

and the corresponding U polynomials are

	$U_0(x+iy)$	$U_1(x+iy)$	$U_2(x+iy)$	$U_3(x+iy)$	$U_4(x+iy)$
Re	1	$2x$	$4x^2-4y^2-1$	$8x^3-24xy^2-4x$	$16x^4-96x^2y^2+16y^4-12x^2+12y^2+1$
Im	0	$2y$	$8xy$	$24x^2y-8y^3-4y$	$64x^3y-64xy^3-24xy$

Applicable when $n=2,3,4,\cdots$ and to both T and U, the recursion formulas

22:11:1 $$\mathrm{Re}\{f_n(x+iy)\}=2x\,\mathrm{Re}\{f_{n-1}(x+iy)\}-2y\,\mathrm{Im}\{f_{n-1}(x+iy)\}-\mathrm{Re}\{f_{n-2}(x+iy)\}$$

and

22:11:2 $$\mathrm{Im}\{f_n(x+iy)\}=2y\,\mathrm{Re}\{f_{n-1}(x+iy)\}+2x\,\mathrm{Im}\{f_{n-1}(x+iy)\}-\mathrm{Im}\{f_{n-2}(x+iy)\}$$

permit any Chebyshev polynomial of complex argument to be constructed.

Inverse Laplace transformation of Chebyshev polynomials leads to the Hermite polynomials [Chapter 24]

22:11:3 $$\int_{\alpha-i\infty}^{\alpha+i\infty}\frac{1}{s^n}T_n\left(1-\frac{1}{s}\right)\frac{\exp(ts)}{2\pi i}ds=\mathfrak{I}\left\{\frac{1}{s^n}T_n\left(1-\frac{1}{s}\right)\right\}=\frac{(-t)^n}{2(2n-1)!t}H_{2n}\left(\sqrt{\frac{t}{2}}\right)$$

and

22:11:4 $$\int_{\alpha-i\infty}^{\alpha+i\infty}\frac{1}{s^{n+1}}U_n\left(\frac{1}{s}-1\right)\frac{\exp(ts)}{2\pi i}ds=\mathfrak{I}\left\{\frac{1}{s^{n+1}}U_n\left(\frac{1}{s}-1\right)\right\}=\frac{t^n}{(2n-1)!\sqrt{2t}}H_{2n-1}\left(\sqrt{\frac{t}{2}}\right)$$

22:12 GENERALIZATIONS: Gegenbauer and Jacobi polynomials

The bivariate Chebyshev polynomial generalizes to the trivariate *Gegenbauer polynomial* (Leopold Bernhard Gegenbauer, Austrian mathematician, 1849 – 1903) or *ultraspherical polynomial* $C_n^{(\lambda)}(x)$. One of several ways of defining this polynomial family is through their generating function

22:12:1 $$\frac{1}{\left(1-2tx+t^2\right)^{\lambda}} = \sum_{n=0}^{\infty} C_n^{(\lambda)}(x)t^n \qquad \lambda \neq 0$$

from which it is follows that the Legendre and Chebyshev polynomials are the special cases

22:12:2 $$P_n(x) = C_n^{(1/2)}(x)$$

22:12:3 $$T_n(x) = \frac{n}{2}\lim_{\lambda\to 0}\left\{\frac{C_n^{(\lambda)}(x)}{\lambda}\right\}$$

and

22:12:4 $$U_n(x) = C_n^{(1)}(x)$$

of the Gegenbauer polynomials. The limiting operation in 22:12:3 is taken as a definition of $C_n^{(0)}(x)$, which is thereby seen to equal $(2/n)T_n(x)$, except $C_0^{(0)}(x)=1$. The $\lambda = 0$ case is not covered by the table.

$C_0^{(\lambda)}(x)$	$C_1^{(\lambda)}(x)$	$C_2^{(\lambda)}(x)$	$C_3^{(\lambda)}(x)$	$C_4^{(\lambda)}(x)$
1	$2\lambda x$	$-\lambda+2\lambda(1+\lambda)x^2$	$2\lambda(1+\lambda)\left[-x+\frac{2}{3}(2+\lambda)x^3\right]$	$\lambda(1+\lambda)\left[\frac{1}{2}-2(2+\lambda)x^2+\frac{2}{3}(2+\lambda)(3+\lambda)x^4\right]$

More expressions may be added through the recursion formula

22:12:5 $$C_n^{(\lambda)}(x) = \frac{2n+2\lambda-2}{n}xC_{n-1}^{(\lambda)}(x) - \frac{n+2\lambda-2}{n}C_{n-2}^{(\lambda)}(x)$$

which applies also to the $\lambda = 0$ case. This recursion formula is the basis of *Equator*'s Gegenbauer polynomial routine (keyword **Cpoly**).

The trivariate Gegenbauer polynomial family is itself a special case

22:12:6 $$C_n^{(\lambda)}(x) = \frac{(2\lambda)_n}{\left(\lambda+\frac{1}{2}\right)_n}P_n^{(\lambda-\frac{1}{2},\lambda-\frac{1}{2})}(x) \qquad \frac{-1}{2} < \lambda \neq 0$$

of the quadrivariate *Jacobi polynomial*. The value of a Jacobi polynomial is determined by: the argument x, a real number with interest concentrated in the $-1 \le x \le 1$ domain; the nonnegative integer degree n; and two distinct real parameters, v and μ. The conventional notation for the Jacobi polynomial is $P_n^{(v,\mu)}(x)$ and its generating function is

22:12:7 $$\frac{2^{v+\mu}}{\sqrt{1-2xt+t^2}\left[1-t+\sqrt{1-2xt+t^2}\right]^{v}\left[1+t+\sqrt{1-2xt+t^2}\right]^{\mu}} = \sum_{n=0}^{\infty} P_n^{(v,\mu)}(x)t^n$$

The Jacobi polynomial is the grandparent of many other orthogonal polynomial families, inasmuch as the Legendre, Chebyshev, Laguerre, associated Laguerre and Hermite polynomials are all special cases:

22:12:8 $$P_n(x) = P_n^{(0,0)}(x)$$

22:12:9 $$T_n(x) = \frac{n!}{\left(\tfrac{1}{2}\right)_n} P_n^{(-1/2,-1/2)}(x)$$

22:12:10 $$U_n(x) = \frac{(n+1)!}{\left(\tfrac{3}{2}\right)_j} P_n^{(1/2,1/2)}(x)$$

22:12:11 $$L_n(x) = \lim_{\mu\to\infty} P_n^{(0,\mu)}\left(1-\frac{2x}{\mu}\right)$$

22:12:12 $$L_n^{(v)}(x) = \lim_{\mu\to\infty} P_n^{(v,\mu)}\left(1-\frac{2x}{\mu}\right)$$

22:12:13 $$H_n(x) = \begin{cases} (-)^{n/2}\, n!!\, L_{n/2}^{(-1/2)}(x^2) & n = 0,2,4,\cdots \\ (-)^{(n-1)/2}\, 2^{n/2}(n-1)!!\, x L_{(n-1)/2}^{(1/2)}(x^2) & n = 1,3,5,\cdots \end{cases}$$

For this reason, many authors prefer to concentrate on the Jacobi polynomials, educing the properties of other orthogonal polynomials therefrom. *Applications* of orthogonal polynomials, on the other hand, are predominantly based on the simpler polynomial families.

Jacobi polynomials are sometimes known as *hypergeometric polynomials* because they may be represented, in two ways, as the product of a binomial coefficient and a Gauss hypergeometric function [Chapters 6 and 60]:

22:12:14 $$P_n^{(v,\mu)}(x) = \binom{n+v}{n} \sum_{j=0}^{\infty} \frac{(-n)_j (n+v+\mu+1)_j}{(1)_j (v+1)_j} \left(\frac{1-x}{2}\right)^j \qquad -3 \le x \le 1$$

22:12:15 $$P_n^{(v,\mu)}(x) = \left(\frac{x+1}{2}\right)^n \binom{n+v}{n} \sum_{j=0}^{\infty} \frac{(-n)_j (-n-\mu)_j}{(1)_j (v+1)_j} \left(\frac{x-1}{x+1}\right)^j \qquad x \ge 0$$

Jacobi polynomials are even/odd according as their degree is even/odd. *Equator*'s Jacobi polynomial routine (keyword **Jacobipoly**) uses $P_0^{(v,\mu)}(x) = 1$ and $P_1^{(v,\mu)}(x) = \frac{1}{2}(v+\mu+2)x + \frac{1}{2}(v-\mu)$ to initialize the recursion formula

22:12:16 $$P_n^{(v,\mu)}(x) = \frac{(2n+v+\mu-1)\left[(2n+v+\mu-2)(2n+v+\mu)x + v^2 - \mu^2\right]}{2n(n+v+\mu)(2n+v+\mu-2)} P_{n-1}^{(v,\mu)}(x) - \frac{(n+v-1)(n+\mu-1)(2n+v+\mu)}{n(n+v+\mu)(2n+v+\mu-2)} P_{n-2}^{(v,\mu)}(x)$$

This formula breaks down when $v + \mu + 1$ equals a negative integer but, nevertheless, *Equator* delivers accurate values of the Jacobi polynomial when the parameters sum to a value close to -2, -3, etc.

For $v > -1$ and $\mu > -1$, the orthogonality of the Jacobi polynomials is manifested by

22:12:17 $$\int_{-1}^{1} (1-t)^v (1+t)^\mu P_m^{(v,\mu)}(t) P_n^{(v,\mu)}(t)\,dt = \begin{cases} 0 & m \ne n \\ \dfrac{2^{v+\mu+1}\Gamma(n+v+1)\Gamma(n+\mu+1)}{(2n+v+\mu+1)n!\,\Gamma(n+v+\mu+1)} & m = n \end{cases}$$

22:13 COGNATE FUNCTIONS: the discrete Chebyshev polynomials

The orthogonality property, addressed in Section 21:14, may be used to construct continuous functions from suitable polynomial families, as is demonstrated in Section 21:15. There is, however, another variety of orthogonality that relates to functions whose values are known only at a set of discrete points, say $y_0, y_1, \cdots, y_j, \cdots, y_J$. These functions, too, are generally polynomials, but the orthogonality relationship that they satisfy is a summation rather than an integration:

22:13:1
$$\sum_{j=0}^{J} \mathrm{w}(y)\Psi_n(y)\Psi_m(y) = \begin{cases} 0 & m \neq n \\ \Omega_n^2 & m = n \end{cases}$$

Such Ψ functions are known as *discrete orthogonal polynomials*.

The simplest family of discrete orthogonal polynomial functions are those with a weight function $\mathrm{w}(y)$ of unity and have been named *discrete Chebyshev polynomials*. The notation $\mathrm{t}_0^{(J)}(y), \mathrm{t}_1^{(J)}(y), \cdots, \mathrm{t}_J^{(J)}(y)$ is used for the $(J+1)$-member set. If y is restricted to the $-1 \le y \le 1$ range, and the data are evenly spaced on this interval, then the orthogonality condition is

22:13:2
$$\sum_{j=0}^{J} \mathrm{t}_m^{(J)}(y_j)\mathrm{t}_n^{(J)}(y_j) = \begin{cases} 0 & m \neq n \\ \Omega_n^2 & m = n \end{cases} \qquad \text{where} \qquad \Omega_n^2 = \frac{(J+1)_{n+1}}{[2n+1](J-n+1)_n}$$

The t polynomials depend on J, as well as on n, and some early members of the family are

$\mathrm{t}_0^{(J)}(y)$	$\mathrm{t}_1^{(J)}(y)$	$\mathrm{t}_2^{(J)}(y)$	$\mathrm{t}_3^{(J)}(y)$	$\mathrm{t}_4^{(J)}(y)$
1	y	$\dfrac{3Jy^2-J-2}{2(J-1)}$	$\dfrac{5J^2y^3-\left(3J^2+6J-4\right)y}{2(J-1)(J-2)}$	$\dfrac{35J^3y^4-10J\left(3J^2+6J-10\right)y^2+3(J+4)(J^2-4)}{8(J-1)(J-2)(J-3)}$

Others can be calculated by the recursion formula

22:13:3
$$\mathrm{t}_n^{(J)}(y) = \frac{(2n-1)Jy\,\mathrm{t}_{n-1}^{(J)}(y)-(n-1)(J+n)\mathrm{t}_{n-2}^{(J)}(y)}{n(J-n+1)}$$

as employed by *Equator*'s discrete Chebyshev polynomial routine (keyword **discCheby**). An application of these polynomials will be found in Section 17:14.

Despite their name, discrete Chebyshev polynomials have less in common with Chebyshev polynomials than they have with Legendre functions, with which they share a unity weight function and to which they reduce as J approaches infinity

22:13:4
$$\lim_{J\to\infty} \mathrm{t}_n^{(J)}(y) = \mathrm{P}_n(y)$$

CHAPTER 23

THE LAGUERRE POLYNOMIALS $L_n(x)$

Laguerre polynomials (Edmond Nicolas Laguerre, French mathematician, 1834−1886) are orthogonal [Section 21:14] on the interval 0 to ∞ with a weight function of $\exp(-x)$. They arise in a number of scientific problems, as in solutions to the wave equation.

23:1 NOTATION

Though the $L_n(x)$ symbol is used for both, Laguerre polynomials are defined in two distinct ways; many authors write $L_n(x)$ for what the *Atlas* represents by $n!L_n(x)$. The name "Laguerre polynomial" may be applied to the general class of orthogonal polynomials discussed in Section 23:12 under the name *associated Laguerre polynomials*.

Though this *Atlas* avoids the ambiguity, $L_n(x)$ is commonly employed to denote the unrelated hyperbolic Struve function [Section 57:13].

23:2 BEHAVIOR

While interest is concentrated on positive arguments, the Laguerre polynomial is defined for all real argument x and all nonnegative integer degree n. As Figure 23-1 shows, the first few Laguerre polynomials are quite simple functions. As n increases, the polynomial displays an oscillatory behavior, the amplitude and period of the oscillations increasing with x, until, beyond r_n, the last zero, the oscillations cease abruptly and the function heads rapidly towards $(-)^n\infty$. Within its oscillatory ambit, the function displays exactly $\mathrm{Int}(n/2)$ minima, $\mathrm{Int}\{(n-1)/2\}$ maxima and n zeros. All the zeros occur between $x = 0$ and an imperfectly known upper value

23:2:1 $$0 < r_1 < r_2 < \cdots < r_m < \cdots < r_n < 2n+1+\sqrt{4n^2+4n+\tfrac{5}{4}} \qquad L_n(r_m) = 0$$

23:3 DEFINITIONS

All orthogonal polynomials, including Laguerre polynomials, can be defined by a generating function

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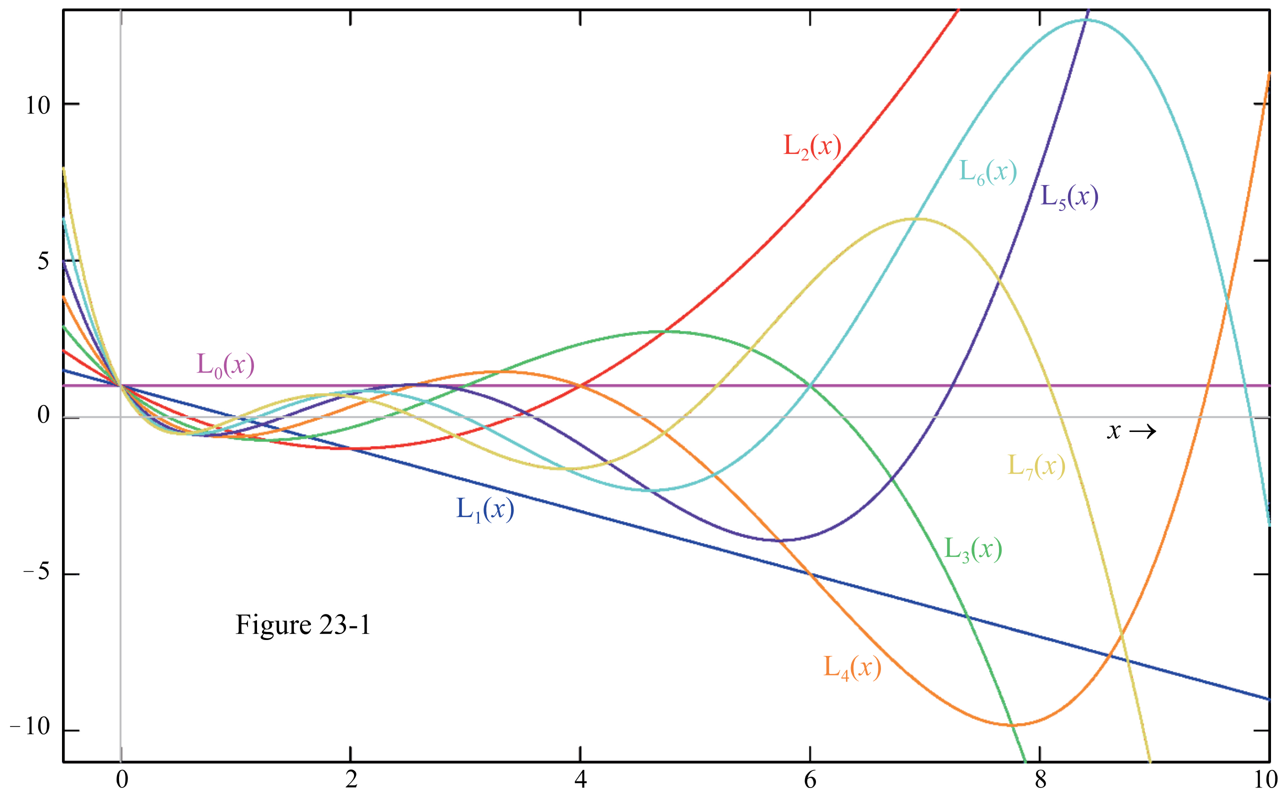
DOI 10.1007/978-0-387-48807-3_24, © Springer Science+Business Media, LLC 2009



23:3:1 $$\frac{1}{1-t}\exp\left(\frac{-xt}{1-t}\right)=\sum_{n=0}^{\infty}\mathrm{L}_n(x)t^n \qquad -1<t<1$$

and by a *Rodrigues's formula*

23:3:2 $$\mathrm{L}_n(x)=\frac{\exp(x)}{n!}\frac{\mathrm{d}^n}{\mathrm{d}x^n}\left\{x^n\exp(-x)\right\}$$

It may also be defined in a manner resembling a binomial expansion:

23:3:3 $$\mathrm{L}_n(x)=\sum_{j=0}^{n}\binom{n}{j}\frac{(-x)^j}{j!}$$

and by formula 22:12:10 as a limiting case of a Jacobi polynomial. An unusual definition, establishing contact with a Bessel function [Chapter 52] is

23:3:4 $$\mathrm{L}_n(x)=\frac{\exp(x)}{n!}\int_0^{\infty}t^n\exp(-t)\mathrm{J}_0\left(2\sqrt{xt}\right)\mathrm{d}t$$

Definitions as $L = K+1 = 2$ hypergeometric functions

23:3:5 $$\mathrm{L}_n(x)=\sum_{j=0}^{\infty}\frac{(-n)_j}{(1)_j(1)_j}x^j$$

23:3:6 $$\mathrm{L}_n(x)=\exp(x)\sum_{j=0}^{\infty}\frac{(1+n)_j}{(1)_j(1)_j}(-x)^j$$

imply the possibility of synthesis [Section 43:14] from an exponential function, thus:

23:3:7
$$\exp(x) \xrightarrow[1]{-n} L_n(x)$$

23:3:8
$$\exp(-x) \xrightarrow[1]{n+1} \exp(-x)L_n(x)$$

One solution of Laguerre's differential equation

23:3:9
$$x\frac{d^2 f}{dx^2} + (1-x)\frac{df}{dx} + nf = 0$$

is

23:3:10
$$f(x) = wL_n(x)$$

where the weight w is arbitrary. The second solution is considerably more complicated [see Murphy].

23:4 SPECIAL CASES

$L_0(x)$	$L_1(x)$	$L_2(x)$	$L_3(x)$	$L_4(x)$	$L_5(x)$
1	$-x+1$	$\frac{1}{2}x^2-2x+1$	$-\frac{1}{6}x^3+\frac{3}{2}x^2-3x+1$	$\frac{1}{24}x^4-\frac{2}{3}x^3+3x^2-4x+1$	$-\frac{1}{120}x^5+\frac{5}{24}x^4-\frac{5}{3}x^3+5x^2-5x+1$

23:5 INTRARELATIONSHIPS

The recursion

23:5:1
$$L_n(x) = \frac{2n-1-x}{n}L_{n-1}(x) - \frac{n-1}{n}L_{n-2}(x) \qquad n = 2,3,4,\cdots$$

relates a Laguerre polynomial to two earlier members of the family. The following argument-addition and argument-multiplication formulas apply to Laguerre polynomial functions:

23:5:2
$$L_n(x+y) = \frac{1}{n!}\left(\frac{-1}{4}\right)^n \sum_{j=0}^{n} \binom{n}{j} H_{2j}\left(\sqrt{x}\right) H_{2n-2j}\left(\sqrt{y}\right)$$

23:5:3
$$L_n(bx) = (1-b)^n \sum_{j=0}^{n} \binom{n}{j} \left(\frac{b}{1-b}\right)^j L_j(x)$$

whereas function-multiplications play a leading role in the *Christoffel-Darboux formula*

23:5:4
$$\sum_{j=0}^{n} L_j(x)L_j(y) = (n+1)\frac{L_n(x)L_{n+1}(y) - L_{n+1}(x)L_n(y)}{x-y}$$

23:6 EXPANSIONS

An explicit expression of the power-series expansion for the Laguerre polynomials is

23:6:1
$$L_n(x) = (-)^n \left[\frac{x^n}{n!} - \frac{n}{1!(n-1)!}x^{n-1} + \frac{n(n-1)}{2!(n-2)!}x^{n-2} - \cdots - (-)^n nx + (-1)^n \right]$$

This expansion is equivalent to 23:3:3.

It is evident from Section 23:4, that the nth Laguerre polynomial $L_n(x)$ is built from a weighted sum of all the powers of x up to x^n. It follows that the first n such expansions can be solved simultaneously to eliminate all the powers $x^0, x^1, x^2, \cdots, x^{n-1}$ and produce an expression for x^n of the form

23:6:2 $$x^n = a_0 L_0(x) + a_1 L_1(x) + \cdots + a_j L_j(x) + \cdots + a_n L_n(x)$$

where the a's are numerical coefficients that can be found by solving the simultaneous equations, or from the integrations

23:6:3 $$a_j = \int_0^\infty x^n \exp(-x) L_j(x)\, dx$$

For example, one finds

23:6:4 $$x^4 = 24L_0(x) - 96L_1(x) + 144L_2(x) - 96L_3(x) + 24L_4(x)$$

The symmetry seen in this example is general. Abramowitz and Stegun [Table 22.10] list coefficients of all such expansions up to that of x^{12}.

23:7 PARTICULAR VALUES

Beyond the first few, no formulas for the exact locations of the zeros of the Laguerre polynomials are known, but see equation 23:9:3. Nor are formulas for the locations or the values of the extrema known to us.

23:8 NUMERICAL VALUES

Equator's Laguerre polynomial routine (keyword **Lpoly**) employs recursion 23:5:1, initiated by $L_0(x) = 1$ and $L_1(x) = 1 - x$.

23:9 LIMITS AND APPROXIMATIONS

For positive argument, the Laguerre polynomials are bounded by

23:9:1 $$-\exp\left(\frac{x}{2}\right) \le L_n(x) \le \exp\left(\frac{x}{2}\right) \qquad x > 0$$

The asymptotic form [Lebedev] adopted by the Laguerre polynomial for large degree is

23:9:2 $$L_n(x) \sim \frac{1}{\sqrt{\pi\sqrt{nx}}} \exp\left(\frac{x}{2}\right) \cos\left(2\sqrt{nx} - \frac{\pi}{4}\right)$$

The mth zero of the Laguerre polynomial is bracketed by

23:9:3 $$\frac{j_{0,m}^2}{4n+2} < r_m < \frac{2m+1}{2n+1}\left[2m + 1 + \sqrt{4m^2 + 4m + \tfrac{5}{4}}\right]$$

where $j_{0,m}$ is the mth zero of the Bessel J_0 function [Section 53:7].

23:10 OPERATIONS OF THE CALCULUS

Differentiation of a Laguerre polynomial generates a difference of two Laguerre polynomials of like argument or, more succinctly, an associated Laguerre polynomial [Section 23:12] of unity order

23:10:1 $$\frac{d}{dx}L_n(x) = \frac{n}{x}\left[L_n(x) - L_{n-1}(x)\right] = -L_{n-1}^{(1)}(x)$$

and the latter iterates to the multiple-differentiation formula

23:10:2 $$\frac{d^m}{dx^m}L_n(x) = (-)^m L_{n-1}^{(m)}(x)$$

An associated Laguerre polynomial of unity order is also generated by integration:

23:10:3 $$\int_0^x L_n(t)\,dt = L_n(x) - L_{n+1}(x) = \frac{x}{n+1}L_{n-1}^{(1)}(x)$$

The orthogonality [Section 22:14] of Laguerre polynomials is established by the integral

23:10:4 $$\int_0^\infty \exp(-t)L_n(t)L_m(t)\,dt = \begin{cases} 0 & m \neq n \\ 1 & m = n \end{cases}$$

See Section 24:15 for an application to numerical integration. Other important integrals include

23:10:5 $$\int_x^\infty \exp(-t)L_n(t)\,dt = \exp(-x)\left[L_n(x) - L_{n-1}(x)\right] = \frac{-n\exp(-x)}{x}L_{n-1}^{(1)}(x)$$

and

23:10:6 $$\int_0^x L_n(t)L_m(x-t)\,dt = \int_0^x L_{n+m}(t)\,dt = \frac{x}{n+m+1}L_{n+m}^{(1)}(x)$$

Many others are listed by Gradshteyn and Ryzhik [Sections 7.41 and 7.42].

Examples of Laplace transforms involving Laguerre polynomials are:

23:10:7 $$\int_0^\infty L_n(t)\exp(-st)\,dt = \mathfrak{L}\{L_n(t)\} = \frac{(s-1)^n}{s^{n+1}}$$

23:10:8 $$\int_0^\infty t^\nu L_n^{(m)}(t)\exp(-st)\,dt = \mathfrak{L}\{t^\nu L_n^{(m)}(t)\} = \frac{\Gamma(\nu+1)}{n!s^{\nu+1}}P_n^{(m,\nu-n-m)}\left(1-\frac{2}{s}\right) \qquad \nu > -1$$

Transform 23:10:8, which generates a Jacobi polynomial, relates to an *associated* Laguerre polynomial [Section 23:12]; merely set $m = 0$ to obtain the formula for a simple Laguerre polynomial.

23:11 COMPLEX ARGUMENT

Applications of Laguerre polynomials generally employ real arguments.

Inverse Laplace transformation of an *associated* Laguerre polynomial generates a Jacobi polynomial.

23:11:1 $$\int_{\alpha-i\infty}^{\alpha+i\infty} \frac{L_n^{(m)}(s)}{s^\nu}\frac{\exp(ts)}{2\pi i}\,ds = \mathfrak{I}\left\{\frac{L_n^{(m)}(s)}{s^\nu}\right\} = \frac{t^{\nu-n-1}(1+t)^n}{\Gamma(\nu)}P_n^{(m,\nu-n-1)}\left(\frac{t-1}{t+1}\right)$$

Set $m = 0$ to obtain the formula for a simple Laguerre polynomial.

23:12 GENERALIZATIONS: including associated Laguerre polynomials

The Laguerre polynomial is the $m = 0$ case of the *associated Laguerre polynomial* $\mathrm{L}_n^{(m)}(x)$

23:12:1 $$\mathrm{L}_n(x) = \mathrm{L}_n^{(0)}(x)$$

Most of the definitions of the Laguerre polynomial, given in Section 23:3, need only minor adjustment to provide a definition of the associated Laguerre polynomial of degree n and order m. For example

23:12:2 $$\frac{1}{(1-t)^{m+1}}\exp\left(\frac{-xt}{1-t}\right) = \sum_{n=0}^{\infty}\mathrm{L}_n^{(m)}(x)t^n \qquad -1 < t < 1$$

23:12:3 $$\mathrm{L}_n^{(m)}(x) = \frac{\exp(x)}{n!x^m}\frac{\mathrm{d}^n}{\mathrm{d}x^n}\left\{x^{n+m}\exp(-x)\right\}$$

23:12:4 $$\mathrm{L}_n^{(m)}(x) = \sum_{j=0}^{n}\binom{n+m}{n-j}\frac{(-x)^j}{j!}$$

23:12:5 $$\mathrm{L}_n^{(\mathrm{m})}(x) = \frac{\exp(x)}{n!x^{m/2}}\int_0^{\infty} t^{(2n+m)/2}\exp(-t)\mathrm{J}_m\left(2\sqrt{tx}\right)\mathrm{d}t$$

23:12:6 $$\mathrm{L}_n^{(m)}(x) = \frac{(m+1)_n}{n!}\sum_{j=0}^{\infty}\frac{(-n)_j}{(1)_j(m+1)_j}x^j$$

and

23:12:7 $$x\frac{\mathrm{d}^2\mathrm{f}}{\mathrm{d}x^2} + (m+1-x)\frac{\mathrm{d}\,\mathrm{f}}{\mathrm{d}x} + n\mathrm{f} = 0 \qquad \text{where} \qquad \mathrm{f}(x) = w\mathrm{L}_n^{(m)}(x)$$

Caution is necessary because the associated Laguerre polynomials are sometimes defined as

23:12:8 $$n!\frac{\mathrm{d}^m}{\mathrm{d}x^m}\mathrm{L}_n(x)$$

This definition leads to properties radically different from those of the polynomials defined by equations 23:12:2−7.

The recursion property, 23:5:1, of the Laguerre polynomial generalizes to

23:12:9 $$\mathrm{L}_n^{(m)}(x) = \frac{2n+m-1-x}{n}\mathrm{L}_{n-1}^{(m)}(x) - \frac{n+m-1}{n}\mathrm{L}_{n-2}^{(m)}(x)$$

which is useful in the construction or extension of the following table, which lists examples of associated Laguerre polynomials.

	$\mathrm{L}_0^{(m)}(x)$	$\mathrm{L}_1^{(m)}(x)$	$\mathrm{L}_2^{(m)}(x)$	$\mathrm{L}_3^{(m)}(x)$	$\mathrm{L}_4^{(m)}(x)$
$m = 0$	1	$-x+1$	$\frac{1}{2}x^2 - 2x + 1$	$-\frac{1}{6}x^3 + \frac{3}{2}x^2 - 3x + 1$	$\frac{1}{24}x^4 - \frac{2}{3}x^3 + 3x^2 - 4x + 1$
$m = 1$	1	$-x+2$	$\frac{1}{2}x^2 - 3x + 3$	$-\frac{1}{6}x^3 + 2x^2 - 6x + 4$	$\frac{1}{24}x^4 - \frac{5}{6}x^3 + 5x^2 - 10x + 5$
$m = 2$	1	$-x+3$	$\frac{1}{2}x^2 - 4x + 6$	$-\frac{1}{6}x^3 + \frac{5}{2}x^2 - 10x + 10$	$\frac{1}{24}x^4 - x^3 + \frac{15}{2}x^2 - 20x + 15$
$m = 3$	1	$-x+4$	$\frac{1}{2}x^2 - 5x + 10$	$-\frac{1}{6}x^3 + 3x^2 - 15x + 20$	$\frac{1}{24}x^4 - \frac{7}{6}x^3 + \frac{21}{2}x^2 - 35x + 35$
$m = 4$	1	$-x+5$	$\frac{1}{2}x^2 - 6x + 15$	$-\frac{1}{6}x^3 + \frac{7}{2}x^2 - 21x + 35$	$\frac{1}{24}x^4 - \frac{4}{3}x^3 + 14x^2 - 56x + 70$

The differential and orthogonality properties of associated Laguerre polynomials are revealed in the following:

23:12:10 $$\frac{d}{dx}L_n^{(m)}(x)=\frac{n}{x}L_n^{(m)}(x)-\frac{n+m}{x}L_{n-1}^{(m)}(x)=-L_{n-1}^{(m+1)}$$

23:12:11 $$\int_0^\infty t^m \exp(-t)L_n^{(m)}(t)L_{n'}^{(m)}(t)\,dt=\begin{cases}0 & n\neq n' \\ (m+n)!/n! & n=n'\end{cases}$$

To this point it has been assumed that the order m of the associated Laguerre polynomials is a nonnegative integer, and this is frequently the case. However, many of the definitions remain valid when m is a negative integer, or a noninteger of either sign, and, indeed, *Equator*'s associated Laguerre polynomial routine (keyword **assocLpoly**), which employs 23:12:9, initialized by $L_0^{(m)}(x)=1$ and $L_1^{(m)}(x)=1+m-x$, does not require m to be an integer. The name *generalized Laguerre polynomial* has been used to describe associated Laguerre polynomials that lack the integer restriction on their order. See equation 45:6:6 for a connection to the incomplete gamma function. Two simple instances may be considered as Hermite polynomials [Chapter 24]:

23:12:12 $$L_n^{(\frac{1}{2})}(x)=\frac{1}{2}\left(\frac{-1}{4}\right)^n\frac{H_{2n+1}\left(\sqrt{x}\right)}{n!\sqrt{x}} \qquad \text{and} \qquad L_n^{(-\frac{1}{2})}(x)=\left(\frac{-1}{4}\right)^n\frac{H_{2n}\left(\sqrt{x}\right)}{n!}$$

The generalized Laguerre polynomial is an instance of the Kummer function [Chapter 47]:

23:12:13 $$L_n^{(\mu)}(x)=\binom{n+\mu}{n}M(-n,\mu+1,x)$$

Yet another generalization arises by allowing the degree of the Laguerre polynomial (*not* the associated version), to adopt noninteger values. The resulting function is no longer a polynomial, being called a *Laguerre function*. It, too, is a simple example of a Kummer function and an $L = K+1 = 2$ hypergeometric function

23:12:14 $$L_\nu(x)=M(-\nu,1,x)=\sum_{j=0}^{\infty}\frac{(-\nu)_j}{(1)_j(1)_j}x^j$$

Confusingly, the name "Laguerre function" is also applied to the solution of the radial portion of the *Schrödinger equation* [Section 46:14] for the hydrogen atom. A good deal more complicated than 23:12:14, this is

23:12:15 $$(n+1)!\exp\left(\frac{-x}{2}\right)x^\ell\frac{d^{2\ell+1}}{dx^{2\ell+1}}L_{n+1}(x)$$

in the terminology of the *Atlas*, n and ℓ being the appropriate quantum numbers.

Perhaps the ultimate generalization of the Laguerre polynomial is when neither the degree nor the order need be an integer. With suitable replacements for factorial functions and Pochhammer polynomials, equations 23:12:3, and 23:12:5−7 can provide definitions of such a *generalized Laguerre function* $L_\nu^{(\mu)}(x)$. It is related to the Kummer function of Chapter 47 and the gamma function of Chapter 43 by

23:12:16 $$L_\nu^{(\mu)}(x)=\frac{\Gamma(1+\nu+\mu)}{\Gamma(1+\nu)\Gamma(1+\mu)}M(-\nu,1+\mu,x)$$

23:13 COGNATE FUNCTIONS

Hermite polynomials, the subject of the next chapter, have many similarities to Laguerre polynomials and there are several linkages between them, including 23:12:12. Both are orthogonal with an exponential function as the weight function; they differ in that the orthogonality interval is semiinfinite for the Laguerre polynomial, but (doubly) infinite for the Hermite version.

23:14 RELATED TOPIC: Fibonacci numbers

In common with many other named polynomial functions, such as the Chebyshev and Hermite polynomials [Chapters 22 and 24], Laguerre polynomials obey a simple three-term recursion – formula 23:5:1 in the Laguerre case. Two other named polynomial functions, not addressed elsewhere in this *Atlas*, obey the similar, but even simpler, three-term recursion

23:14:1 $$\mathrm{f}_{n+1}(x) = x\,\mathrm{f}_n(x) + \mathrm{f}_{n-1}(x)$$

These are the Lucas polynomials and the Fibonacci polynomials, but only the latter will be considered here.

We adopt the symbol $\mathrm{Fib}_n(x)$ for *Fibonacci polynomials*, which are defined by recursion 23:14:1 initialized by $\mathrm{f}_0(x) = 0$, $\mathrm{f}_1(x) = 1$. Early members are

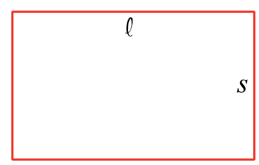
$\mathrm{Fib}_0(x)$	$\mathrm{Fib}_1(x)$	$\mathrm{Fib}_2(x)$	$\mathrm{Fib}_3(x)$	$\mathrm{Fib}_4(x)$	$\mathrm{Fib}_5(x)$	$\mathrm{Fib}_6(x)$	$\mathrm{Fib}_7(x)$	$\mathrm{Fib}_8(x)$
0	1	x	x^2+1	x^3+2x	x^4+3x^2+1	x^5+4x^3+3x	$x^6+5x^4+6x^2+1$	$x^7+6x^5+10x^3+4x$

Initialized by $\mathrm{Fib}_0(x)$ and $\mathrm{Fib}_1(x)$, recursion 23:14:1 is used by *Equator*'s Fibonacci polynomial routine, keyword **Fibpoly**.

Fibonacci numbers, are the $x = 1$ values of the eponymous polynomial, each such number being the sum of its two predecessors. Important in a variety of contexts, these numbers even appear in Dan Brown's popular novel *The Da Vinci Code*. They are named for Leonardo Fibonacci ("Leonardo of Pisa" 1175 – 1230), but were known to the ancients in connection with the *golden section* [Newman, page 98], which is the positive number υ that exceeds its reciprocal by unity, that is

23:14:2 $$\upsilon = \frac{1+\upsilon}{\upsilon} \qquad \text{whence} \qquad \upsilon = \frac{\sqrt{5}+1}{2} = 1.6180\,33988\,74989$$

The "golden" sobriquet has its origin in the claim that a rectangle has the most aesthetically appealing shape when the ratio ℓ/s of the longer to the shorter side equals the ratio of the semiperimeter to the longer side, so that



23:14:3 $$\frac{\ell}{s} = \frac{\ell + s}{\ell} = \upsilon$$

More obscurely, υ is the radius of the circle that exscribes a regular decagon of unity side.

The ratio $\mathrm{Fib}(n)/\mathrm{Fib}(n-1)$ of a Fibonacci number to its predecessor is alternately less than and greater than the golden section and this ratio converges rapidly to υ. This convergence is incorporated into the exact formula

23:14:4 $$\mathrm{Fib}(n) = \mathrm{Fib}_n(1) = \mathrm{Round}\left(\frac{\upsilon^n}{\sqrt{5}}\right) \qquad n = 0,1,2,\cdots$$

This is how *Equator*'s Fibonacci number routine (keyword **Fibnum**) primarily operates, generating exact integers for values of n through 73. For $74 \le n \le 1476$, a 15-digit approximation is provided. The first eighteen Fibonacci numbers are

n	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
$\mathrm{Fib}(n)$	0	1	1	2	3	5	8	13	21	34	55	89	144	233	377	610	987	1597

CHAPTER 24

THE HERMITE POLYNOMIALS $H_n(x)$

Named for the Frenchman, Charles Hermite (1822 − 1901) these polynomials are orthogonal [Section 21:14] on the infinite interval $-\infty < x < \infty$ with a weight function of $\exp(-x^2)$. They arise in physics, as in the solution of Schrödinger's differential equation for a simple harmonic oscillator [Section 24:13]. This differential equation belongs to a broad class of second-order differential equations, many members of which may be solved by the approach exposed in Section 24:14.

24:1 NOTATION

The notation $H_n(x)$ is in general use for the Hermite polynomial of degree n and argument x, however it is occasionally defined with sign opposite to that adopted here. An *alternative Hermite function* is also in common use. Unfortunately it, too, is sometimes symbolized $H_n(x)$, but more often $He_n(x)$ is preferred. The *Atlas* uses the $H_n(x)$ variant exclusively. The relationship between the two varieties of Hermite function is

24:1:1 $$\mathrm{He}_n(x) = \frac{1}{2^{n/2}} \mathrm{H}_n\left(\frac{x}{\sqrt{2}}\right)$$

The conventional notation for the unrelated Struve function [Chapter 57] is also $H_n(x)$. To avoid ambiguity, the *Atlas* denotes the Struve function by $h_n(x)$.

24:2 BEHAVIOR

The Hermite polynomial $H_n(x)$ has exactly n zeros, $\mathrm{Int}(n/2)$ minima and $\mathrm{Int}((n-1)/2)$ maxima. These features are symmetrically disposed about $x = 0$, and all occur in the zone $-\sqrt{2n} < x < \sqrt{2n}$. Outside this domain $|H_n(x)|$ increases monotonically, except for $H_0(x)$, remaining bounded globally by

24:2:1 $$\left|\mathrm{H}_n(x)\right| \le \sqrt{2^n n! \exp(x^2)}$$

The range spanned by the oscillations of $H_n(x)$ increases so dramatically with increasing n that Figure 24-1 portrays the behavior of $H_n(x)/n!$ rather than that of the Hermite polynomials themselves.

K.B. Oldham et al., *An Atlas of Functions, Second Edition*,
DOI 10.1007/978-0-387-48807-3_25,

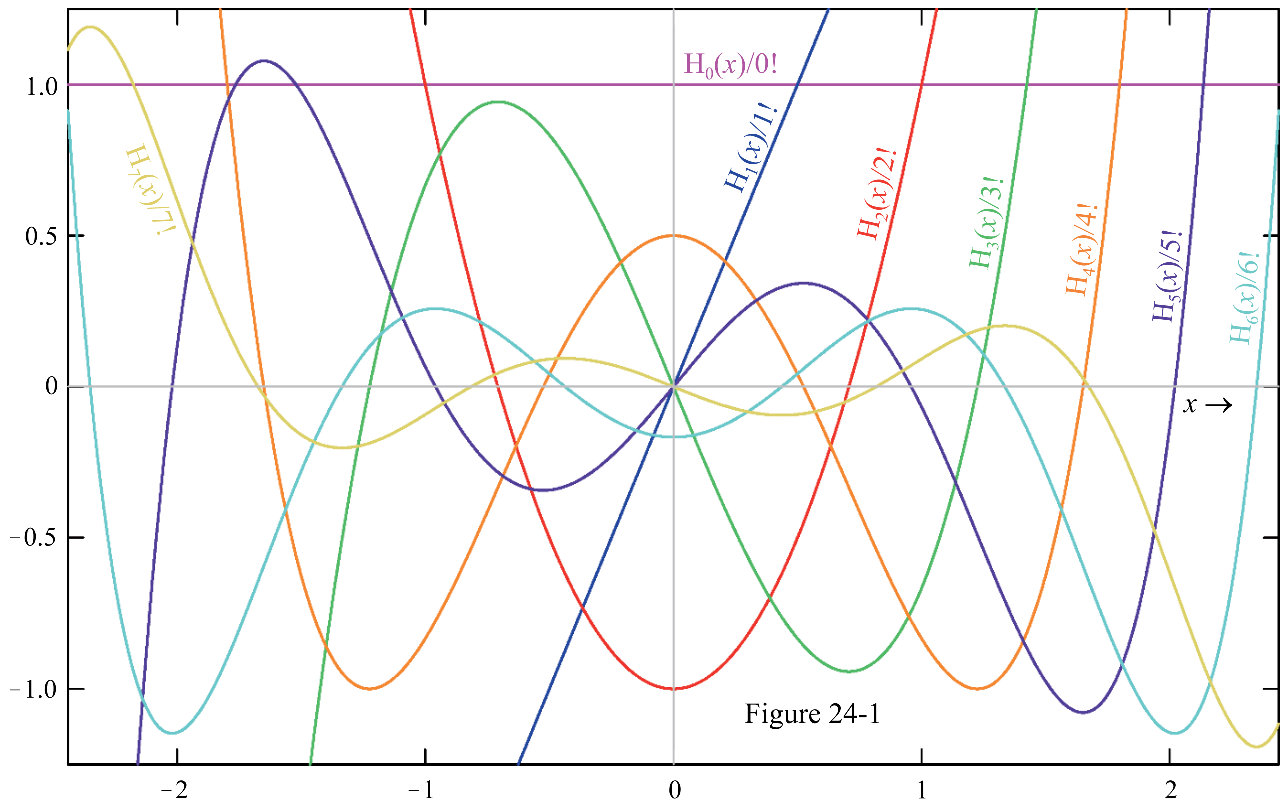


Figure 24-1

24:3 DEFINITIONS

In common with all orthogonal polynomials, there is a generating function

24:3:1
$$\exp\left(2xt-t^2\right)=\sum_{n=0}^{\infty}\mathrm{H}_n(x)\frac{t^n}{n!}$$

and a *Rodrigues's formula*

24:3:2
$$\mathrm{H}_n(x)=(-)^n\exp\left(x^2\right)\frac{\mathrm{d}^n}{\mathrm{d}x^n}\exp\left(-x^2\right)$$

that can serve as definitions of the Hermite polynomial. As well, it can be defined as the limiting operation 22:12:12 applied to the associated Laguerre polynomial, or through the integral representation

24:3:3
$$\mathrm{H}_n(x)=\frac{2^{n+1}}{\sqrt{\pi}}\exp(x^2)\int_0^{\infty}t^n\exp(-t^2)\cos\left(2xt-\frac{n\pi}{2}\right)\mathrm{d}t$$

There are two formulas that go by the name of *Hermite's differential equation*. One is

24:3:4
$$\frac{\mathrm{d}^2\mathrm{f}}{\mathrm{d}x^2}-2x\frac{\mathrm{d}\mathrm{f}}{\mathrm{d}x}+2n\mathrm{f}=0$$

and it is solved by $\mathrm{f}=w\mathrm{H}_n(x)$, w being an arbitrary constant. The second version lacks both of the 2's present in 24:3:4, and its solution is the function defined in 24:1:1.

Routes by which a Hermite polynomial may be synthesized [Section 43:14] are

24:3:5 $$\exp(x) \xrightarrow[1/2]{-n/2} \frac{H_n\left(\sqrt{x}\right)}{(-2)^{n/2}(n-1)!!} \qquad n = 0,2,4,\cdots$$

and

24:3:6 $$\exp(x) \xrightarrow[3/2]{(1-n)/2} \frac{-H_n\left(\sqrt{x}\right)}{(-2)^{(n+1)/2}n!!\sqrt{x}} \qquad n = 1,3,5,\cdots$$

Expansions 24:6:4 and 24:6:5 lie at the heart of these syntheses.

24:4 SPECIAL CASES

$H_0(x)$	$H_1(x)$	$H_2(x)$	$H_3(x)$	$H_4(x)$	$H_5(x)$	$H_6(x)$
1	$2x$	$4x^2-2$	$8x^3-12x$	$16x^4-48x^2+12$	$32x^5-160x^3+120x$	$64x^6-480x^4+720x^2-120$

24:5 INTRARELATIONSHIPS

The reflection formula

24:5:1 $$H_n(-x) = (-)^n H_n(x)$$

establishes that a Hermite polynomial is even or odd in accordance with the parity of its degree. The recursion formula

24:5:2 $$H_n(x) = 2x H_{n-1}(x) - (2n-2) H_{n-2}(x) \qquad n = 2,3,4,\cdots$$

links three consecutive members of the family.

There is an argument-addition formula for Hermite polynomials

24:5:3 $$H_n(x+y) = \frac{1}{2^{n/2}} \sum_{j=0}^{n} \binom{n}{j} H_j\left(\sqrt{2}x\right) H_{n-j}\left(\sqrt{2}y\right)$$

and also an example of the *Christoffel-Darboux formulas*

24:5:4 $$\sum_{j=0}^{n} \frac{H_j(x)H_j(y)}{2^j j!} = \frac{H_{n+1}(x)H_n(y) - H_n(x)H_{n+1}(y)}{2^{n+1}n!(x-y)}$$

Two infinite series of Hermite polynomials have the following sums:

24:5:5 $$\frac{H_0(x)}{0!} - \frac{H_2(x)}{2!} + \frac{H_4(x)}{4!} - \cdots = e\cos(2x)$$

24:5:6 $$\frac{H_1(x)}{1!} - \frac{H_3(x)}{3!} + \frac{H_5(x)}{5!} - \cdots = e\sin(2x)$$

where e is the base of natural logarithms [Chapter 1]. Series akin to these in which the signs are uniformly positive sum to $(1/e)\cosh(2x)$ and $(1/e)\sinh(2x)$ respectively and consequently

24:5:7 $$\frac{H_0(x)}{0!} + \frac{H_1(x)}{1!} + \frac{H_2(x)}{2!} + \cdots = \exp(2x-1)$$

which, alternatively, follows directly from the generating function 24:3:1.

24:6 EXPANSIONS

The power-series expansion of the Hermite polynomials can be written in several equivalent ways:

24:6:1 $$H_n(x) = (2x)^n - \frac{n(n-1)}{1!}(2x)^{n-2} + \frac{n(n-1)(n-2)(n-3)}{2!}(2x)^{n-4} - \cdots \begin{cases} +(-2)^{n/2}(n-1)!! & n = 0,2,4,\cdots \\ -(-2)^{(n+1)/2}n!!x & n = 1,3,5,\cdots \end{cases}$$

24:6:2 $$H_n(x) = n!\sum_{j=0}^{\mathrm{Int}(n/2)} \frac{(-1)^j}{j!(n-2j)!}(2x)^{n-2j}$$

24:6:3 $$H_n(x) = (2x)^n \sum_{j=0}^{\mathrm{Int}(n/2)} \frac{\left(\frac{-n}{2}\right)_j \left(\frac{1-n}{2}\right)_j}{(1)_j}\left(\frac{-1}{x^2}\right)^j$$

24:6:4 $$H_n(x) = (-2)^{n/2}(n-1)!!\sum_{j=0}^{n/2} \frac{\left(\frac{-n}{2}\right)_j}{\left(\frac{1}{2}\right)_j (1)_j} x^{2j} \qquad n = 0,2,4,\cdots$$

24:6:5 $$H_n(x) = -(-2)^{(n+1)/2} n!! \sum_{j=0}^{(n-1)/2} \frac{\left(\frac{1-n}{2}\right)_j}{(1)_j \left(\frac{3}{2}\right)_j} x^{2j+1} \qquad n = 1,3,5,\cdots$$

24:7 PARTICULAR VALUES

At $x = 0$, Hermite polynomials of even degree have an integer value, while those of odd degree are zero:

24:7:1 $$H_n(0) = \begin{cases} (-2)^{n/2}(n-1)!! & n = 0,2,4,\cdots \\ 0 & n = 1,3,5,\cdots \end{cases}$$

Apart from these, formulas for the zeros are known only for the lowest degrees. The two zeros of $H_2(x)$ are $\pm 1/\sqrt{2}$, the three zeros of $H_3(x)$ are 0, $\pm\sqrt{3/2}$, and the four zeros of $H_4(x)$ are

24:7:2 $$H_4(r) = 0 \qquad r = \pm\sqrt{\frac{3 \pm \sqrt{6}}{2}} = \begin{cases} \pm 0.52464\ 76232\ 75290 \\ \pm 1.6506\ 80123\ 88578 \end{cases}$$

Because of the simplicity of formula 24:10:1, these zeros of H_4 are the locations of extrema of H_5 and inflections of H_6.

24:8 NUMERICAL VALUES

Equator's Hermite polynomial routine (keyword **Hpoly**) uses recursion 24:5:2, initialized by $H_0(x) = 1$ and $H_1(x) = 2x$, to calculate exact values of $H_n(x)$.

24:9 LIMITS AND APPROXIMATIONS

A Hermite polynomial of large degree is approximated by the oscillatory function

24:9:1 $$\mathrm{H}_n(x) \approx \begin{cases} (-2)^{n/2}(n-1)!!\exp\left(x^2/2\right)\cos\left(\sqrt{2n+1}\;x\right) & \text{even } n\to\infty \\ (-2)^{(n-1)/2}n!!\exp\left(x^2/2\right)\sin\left(\sqrt{2n+1}\;x\right) & \text{odd } n\to\infty \end{cases}$$

This approximation is best for small $|x|$ and fails hopelessly for $|x| > \sqrt{2n}$, where the polynomial ceases to oscillate. Approximation 24:9:1 predicts that the zeros of the Hermite polynomial lie at

24:9:2 $$r \approx \frac{j\pi}{2\sqrt{2n+1}} \quad \text{where} \begin{cases} j = \pm1, \pm3, \pm5, \cdots & \text{if } n \text{ large and even} \\ j = 0, \pm2, \pm4, \cdots & \text{if } n \text{ large and odd} \end{cases}$$

As expected, this approximation has least error for zeros close to $x = 0$.

24:10 OPERATIONS OF THE CALCULUS

The following simple results hold:

24:10:1 $$\frac{\mathrm{d}}{\mathrm{d}x}\mathrm{H}_n(x) = 2n\,\mathrm{H}_{n-1}(x) \qquad n = 1,2,3,\cdots$$

24:10:2 $$\frac{\mathrm{d}}{\mathrm{d}x}\left\{\exp(-x^2)\,\mathrm{H}_n(x)\right\} = -\exp(-x^2)\,\mathrm{H}_{n+1}(x)$$

24:10:3 $$\int_0^x \mathrm{H}_n(t)\,\mathrm{d}t = \begin{cases} \dfrac{\mathrm{H}_{n+1}(x)}{2n+2} & n = 0,2,4,\cdots \\ \dfrac{\mathrm{H}_{n+1}(x) - (-2)^{(n+1)/2}\,n!!}{2n+2} & n = 1,3,5,\cdots \end{cases}$$

and

24:10:4 $$\int_0^x \exp\left(-t^2\right)\mathrm{H}_n(t)\,\mathrm{d}t = \begin{cases} -\exp\left(-x^2\right)\mathrm{H}_{n-1}(x) & n = 0,2,4,\cdots \\ (-2)^{(n-1)/2}(n-2)!! - \exp\left(-x^2\right)\mathrm{H}_{n-1}(x) & n = 1,3,5,\cdots \end{cases}$$

Among the many listed by Gradshteyn and Ryzhik [Sections 7.37 and 7.38], the following definite integrals are of particular interest

24:10:5 $$\int_{-\infty}^{\infty} \exp(-t^2)\mathrm{H}_n(bt)\,\mathrm{d}t = \begin{cases} \sqrt{\pi}\,n!(b^2-1)^{n/2}/(\tfrac{n}{2})! & n = 0,2,4,\cdots \\ 0 & n = 1,3,5,\cdots \end{cases}$$

24:10:6 $$\int_{-\infty}^{\infty} t\exp(-t^2)\mathrm{H}_n(bt)\,\mathrm{d}t = \begin{cases} 0 & n = 0,2,4,\cdots \\ \sqrt{\pi}\,n!!b(2b^2-2)^{(n-1)/2} & n = 1,3,5,\cdots \end{cases}$$

24:10:7 $$\int_{-\infty}^{\infty} \exp\left(-(t-x)^2\right)\mathrm{H}_n(t)\,\mathrm{d}t = \begin{cases} \sqrt{\pi}\,(2x)^n & n = 2,4,6,\cdots \\ 0 & n = 1,3,5,\cdots \end{cases}$$

24:10:8 $$\int_{-\infty}^{\infty} t^n\exp(-t^2)\mathrm{H}_n(bt)\,\mathrm{d}t = \sqrt{\pi}\,n!\mathrm{P}_n(b)$$

24:10:9 $$\int_0^\infty x^\nu \exp\left(\frac{-x^2}{\mu^2}\right)\mathrm{H}_n(x)\,\mathrm{d}x = \begin{cases} \dfrac{(-4)^{n/2}}{2}\left(\dfrac{1}{2}\right)_{n/2}\Gamma\left(\dfrac{1+\nu}{2}\right)\mu^{1+\nu}\,\mathrm{F}\left(\dfrac{-n}{2},\dfrac{1+\nu}{2},\dfrac{1}{2},\mu^2\right) & \nu>-1 \quad n=2,4,6,\cdots \\ (-4)^{(n-1)/2}2\left(\dfrac{1}{2}\right)_{(n-1)/2}\Gamma\left(\dfrac{1+\nu}{2}\right)\mu^{2+\nu}\,\mathrm{F}\left(\dfrac{1-n}{2},1+\dfrac{\nu}{2},\dfrac{3}{2},\mu^2\right) & \nu>-2 \quad n=1,3,5,\cdots \end{cases}$$

The P, Γ, and F functions in the previous two equations are the Legendre polynomial [Chapter 21], the gamma function [Chapter 43] and the Gauss hypergeometric function [Chapter 60].

The orthogonality of Hermite polynomials is established by

24:10:10 $$\int_{-\infty}^{\infty}\exp(-t^2)\mathrm{H}_n(t)\mathrm{H}_m(t)\,\mathrm{d}t = \begin{cases} 0 & m\neq n \\ \sqrt{\pi}\,2^n n! & m=n \end{cases}$$

Term-by term Laplace transformation of formulas 24:6:5 and 24:6:6 leads to

24:10:11 $$\mathfrak{L}\{\mathrm{H}_n(t)\} = \begin{cases} \dfrac{(-2)^{n/2}(n-1)!!}{s}\displaystyle\sum_{j=0}^{n/2}\left(\tfrac{-n}{2}\right)_j\left(\dfrac{4}{s^2}\right)^j = \dfrac{8^{n/2}(n-1)!!\frac{n}{2}!}{s^{n+1}}\mathrm{e}_{n/2}\left(\dfrac{-s^2}{4}\right) & n=0,2,4,\cdots \\ \dfrac{-(-2)^{(n+1)/2}n!!}{s^2}\displaystyle\sum_{j=0}^{(n-1)/2}\left(\tfrac{1-n}{2}\right)_j\left(\dfrac{4}{s^2}\right)^j = \dfrac{8^{(n+1)/2}n!!\frac{n-1}{2}!}{s^{n+1}}\mathrm{e}_{(n-1)/2}\left(\dfrac{-s^2}{4}\right) & n=1,3,5,\cdots \end{cases}$$

in which $\mathrm{e}_n(x)$ represents the exponential polynomial [Section 26:13].

24:11 COMPLEX ARGUMENT

Applications of Hermite polynomials with complex argument are rare.

An example of an inverse Laplace transform involving a Hermite polynomial is

24:11:1 $$\int_{\alpha-i\infty}^{\alpha+i\infty}\frac{\mathrm{H}_{2n}\left(\sqrt{s}\right)}{s^{(2n+1)/2}}\frac{\exp(ts)}{2\pi i}\,\mathrm{d}s = \mathfrak{I}\left\{\frac{\mathrm{H}_{2n}\left(\sqrt{s}\right)}{s^{(2n+1)/2}}\right\} = \frac{4^n}{\sqrt{\pi t}}(1+t)^n\,\mathrm{T}_n\left(\frac{1-t}{1+t}\right)$$

24:12 GENERALIZATIONS

Hermite polynomials are special cases of the parabolic cylinder function [Chapter 46]. Those parabolic cylinder functions whose order ν is a nonnegative integer n are related through

24:12:1 $$\mathrm{D}_n(x) = \frac{1}{2^{n/2}}\exp\left(\frac{-x^2}{4}\right)\mathrm{H}_n\left(\frac{x}{\sqrt{2}}\right)$$

to the corresponding Hermite polynomial. The parabolic cylinder function itself generalizes to the Kummer and Tricomi functions [Chapters 47 and 48].

24:13 COGNATE FUNCTIONS: the Hermite functions

By analogy with other orthogonal polynomials, one might expect that the Hermite function would be $\mathrm{H}_\nu(x)$ and arise by relaxing the requirement that the degree of the Hermite polynomial be an integer. This is not so. The name *Hermite function of degree n* is actually given to the product

24:13:1 $$f_n(x) = (-)^n \exp\left(-x^2/2\right) H_n(x) \qquad n = 0,1,2,\cdots$$

This function is the Hermite polynomial, multiplied by the square root of its weight function, so that it satisfies an orthogonality relationship without any further weighting:

24:13:2 $$\int_{-\infty}^{\infty} f_n(t) f_m(t)\,dt = \begin{cases} 0 & m \neq n \\ 2^n \sqrt{\pi}\, n! & m = n \end{cases}$$

The Hermite function may be defined by the repeated derivative

24:13:3 $$f_n(x) = \exp\left(\frac{x^2}{2}\right) \frac{d^n}{dx^n} \exp\left(-x^2\right) \qquad n = 0,1,2,\cdots$$

or, in operator notation [Boas, Section 12.22], as

24:13:4 $$f_n(x) = \left(\frac{d}{dx} - x\right)^n \exp\left(\frac{-x^2}{2}\right) \qquad n = 0,1,2,\cdots$$

For the harmonic oscillator, it satisfies the *Schrödinger equation*

24:13:5 $$\frac{d^2 f}{dx^2} + (2n + 1 - x^2) f = 0$$

of atomic physics, and is related to the parabolic cylinder function of Chapter 46 by

24:13:6 $$f_n(x) = 2^{n/2} \exp\left(x^2/2\right) D_n\left(\sqrt{2}\, x\right)$$

24:14 RELATED TOPIC: solving differential equations

Equation 24:3:4 and 24:13:5 are examples drawn from the large and important class of differential equations that may be written in the general form

24:14:1 $$\frac{d^2}{dx^2} f(x) - 2B(x) \frac{d}{dx} f(x) + C(x) f(x) = 0$$

The algebraic functions B and C may or may not involve the independent variable x and may even be zero. For reasons that need not detain us, such a differential equation is described as linear, homogeneous, second-order, and ordinary. It, and its inhomogeneous counterpart, to be considered later, are the only differential equations that will be addressed in this section. The solution is invariably of the form

24:14:2 $$f = w_1 f_1(x) + w_2 f_2(x)$$

where w_1 and w_2 are arbitrary weighting factors and f_1, f_2 are functions either of which separately satisfy the differential equation. To provide a complete solution, f_1 and f_2 must be "linearly independent", a condition that will be identified later.

There are many approaches to solving 24:14:1 [see Murphy for a concise description]. One that has merit requires that the so-called *invariant function*, defined by

24:14:3 $$I(x) = C(x) - B^2(x) + \frac{d}{dx} B(x)$$

first be determined. Then the solution of the differential equation 24:14:1 is

24:14:4 $$f = w_1 f_1(x) + w_2 f_2(x) = \exp\left(\int B(x)\,dx\right)\left[w_1 F_1(x) + w_2 F_2(x)\right]$$

where the integration is indefinite and where the identities of the functions F_1 and F_2 depend solely on the invariant function I. The table that follows gives the two F functions (or sometimes just one of them) that correspond to the listed standard forms of the invariant I(x) function.. The numbers in brackets reference the relevant chapter or section. Do not make the easy mistake of regarding the F's as the solutions to the differential equation: they require multiplication by $\exp\left(\int B(x)\,dx\right)$ first, which often simplifies the expression.

I(x)	$F_{1,2}(x)$	Ref.
$-4a^2x^{p-2}$	$\sqrt{x}\,I_{\pm\frac{1}{p}}\left(\frac{a}{p}x^{p/2}\right)$ or, if $1/p$ is the integer n, $\sqrt{x}\,I_n\left(anx^{1/2n}\right)$, $\sqrt{x}\,K_n\left(anx^{1/2n}\right)$	[50]
$4a^2x^{p-2}$	$\sqrt{x}\,J_{\pm\frac{1}{p}}\left(\frac{a}{p}x^{p/2}\right)$ or, if $1/p$ is the integer n, $\sqrt{x}\,J_n\left(anx^{1/2n}\right)$, $\sqrt{x}\,Y_n\left(anx^{1/2n}\right)$	[53]
$-64a^2x^2$	$\sqrt{x}\,I_{\pm\frac{1}{4}}\left(ax^2\right)$	[50]
$(v+½)a^2-(a^4x^2/4)$	$D_v(\pm ax)$	[46]
$\frac{(2v+1)a^2-x^2}{a^4}$	$\exp\left(\frac{-x^2}{2a^2}\right)M\left(\frac{-v}{2},\frac{1}{2},\frac{x^2}{a^2}\right)$, $x\exp\left(\frac{-x^2}{2a^2}\right)M\left(\frac{1-v}{2},\frac{3}{2},\frac{x^2}{a^2}\right)$	[47]
$(2n+1)a^2-a^4x^2$	$\exp(-a^2x^2/2)H_n(ax)$, $\exp\left(-a^2x^2/2\right)M\left(-n/2,1/2,a^2x^2\right)$	[24]
$-(2n+1)a^2-a^4x^2$	$\exp(a^2x^2/2)i^n\mathrm{erfc}(\pm ax)$	[40:13]
$64a^2x^2$	$\sqrt{x}\,J_{\pm\frac{1}{4}}\left(ax^2\right)$	[53]
a^3x	$\mathrm{Ai}(ax)$, $\mathrm{Bi}(ax)$	[56]
a^3x+a^2b	$\mathrm{Ai}(ax+b)$, $\mathrm{Bi}(ax+b)$	[56]
$-a^2$	$\exp(\pm ax)$	[26]
0	1, x	
a^2	$\sin(ax)$, $\cos(ax)$	[32]
$a^2/4x$	$\sqrt{x}\,J_1\left(a\sqrt{x}\right)$, $\sqrt{x}\,Y_1\left(a\sqrt{x}\right)$	[52,54]
$-a^2/4x$	$\sqrt{x}\,I_1\left(a\sqrt{x}\right)$, $\sqrt{x}\,K_1\left(a\sqrt{x}\right)$	[49,51]
$(¼-a^2)/x^2$	$x^{½\pm a}$	[12]
$1/4x^2$	$\sqrt{x}$, $\sqrt{x}\ln(x)$	[25]
$(¼+a^2)/x^2$	$\sqrt{x}\sin(a\ln x)$, $\sqrt{x}\cos(a\ln x)$	[32]
$a^2+(1/4x^2)$	$\sqrt{x}\,J_0(ax)$, $\sqrt{x}\,Y_0(ax)$	[52,54]
$-a^2+(1/4x^2)$	$\sqrt{x}\,I_0(ax)$, $\sqrt{x}\,K_0(ax)$	[49,51]
$a^2-\frac{\nu^2-\frac{1}{4}}{x^2}$	$\sqrt{x}\,J_{\pm\nu}(ax)$ or, if ν is the integer n $\sqrt{x}\,J_n(ax)$, $\sqrt{x}\,Y_n(ax)$	[52-54]

$I(x)$	$F_{1,2}(x)$	Ref.
$-a^2-\dfrac{\nu^2-\frac{1}{4}}{x^2}$	$\sqrt{x}\,I_{\pm\nu}(ax)$ or, if ν is the integer n $\sqrt{x}\,I_n(ax)$, $\sqrt{x}\,K_n(ax)$	[49,51]
$\dfrac{-ax-\nu^2+1}{4x^2}$	$\sqrt{x}\,I_\nu\left(\sqrt{ax}\right)$, $\sqrt{x}\,K_\nu\left(\sqrt{ax}\right)$	[50,51]
$\dfrac{3a^2-(8n+4)ax-4x^2}{16a^2x^2}$	$x^{1/4}\exp\left(\dfrac{x}{2a}\right)\mathrm{i}^n\mathrm{erfc}\left(\pm\sqrt{\dfrac{x}{a}}\right)$	[40:13]
$[1+(4n+2)ax-a^2x^2]/4x^2$	$\sqrt{x}\exp(-ax/2)L_n(ax)$	[23]
$\dfrac{1-4\mu^2+4\nu ax-a^2x^2}{4x^2}$	$M_{\nu,\mu}(ax)$, $W_{\nu,\mu}(ax)$	[48:13]
$-2/(a^2+x^2)$	(a^2+x^2), $ax+(a^2+x^2)\arctan(x/a)$	[35]
$\dfrac{-2a}{x(a-x)^2}$	$\dfrac{x}{a-x}$, $\dfrac{a+x}{a}+\dfrac{2x}{a-x}\ln\left(\dfrac{x}{a}\right)$	[25]
$\dfrac{a^2+3x^2}{4x^2(a^2-x^2)}$	$\sqrt{x}\,E\left(\dfrac{x}{a}\right)$, $\sqrt{x}\left[E\left(\dfrac{\sqrt{a^2-x^2}}{a}\right)-K\left(\dfrac{\sqrt{a^2-x^2}}{a}\right)\right]$	[61]
$\dfrac{\nu^2+\nu+1}{a^2-x^2}+\dfrac{x^2}{\left(a^2-x^2\right)^2}$	$\sqrt{a^2-x^2}\,P_\nu\left(\dfrac{x}{a}\right)$, $\sqrt{a^2-x^2}\,Q_\nu\left(\dfrac{x}{a}\right)$ polynomials if $\nu = n$	[59]
$\dfrac{(n^2+\frac{1}{2})}{a^2-x^2}+\dfrac{\frac{3}{4}x^2}{\left(a^2-x^2\right)^2}$ $n\neq 0$	$(a^2-x^2)^{1/4}\,T_n\left(\dfrac{x}{a}\right)$, $(a^2-x^2)^{3/4}\,U_{n-1}\left(\dfrac{x}{a}\right)$	[22]
$\dfrac{a^4+2a^2x^2-15x^4}{4x^2(a^2-x^2)^2}$	$\sqrt{x(a^2-x^2)}\,K\left(\dfrac{x}{a}\right)$, $\sqrt{x(a^2-x^2)}\,K\left(\dfrac{\sqrt{a^2-x^2}}{a}\right)$	[61]
$\dfrac{n^2+2n\lambda+\lambda+\frac{1}{2}}{a^2-x^2}+\dfrac{(\frac{3}{4}+\lambda-\lambda^2)x^2}{\left(a^2-x^2\right)^2}$	$\left(a^2-x^2\right)^{(1+2\lambda)/4}C_n^{(\lambda)}\left(\dfrac{x}{a}\right)$	[22:12]
$-\dfrac{(\alpha+\beta)^2-1}{4(a-x)^2}-\dfrac{\alpha\beta}{x(a-x)}-\dfrac{(4\gamma^2-1)a}{4x^2(a-x)}$	$x^{\frac{1}{2}\pm\gamma}\left(a-x\right)^{(\alpha+\beta+1)/2}F\left(\alpha+\frac{1}{2}\pm\gamma,\beta+\frac{1}{2}\pm\gamma;1\pm 2\gamma;\dfrac{x}{a}\right)$	[60]
a^2/x^4	$x\exp(\pm a/x)$	[27]
$(-a^2-2x^2)/x^4$	$x(x\pm a)\exp(\mp a/x)$	[27]
$(a^2-2x^2)/x^4$	$x^2\cos(a/x)+ax\sin(a/x)$, $x^2\sin(a/x)-ax\cos(a/x)$	[32]

By setting $B(x)$ equal to zero, one can see that F_1 and F_2 are nothing but solutions of the differential equation

24:14:5 $$\frac{d^2}{dx^2}F(x)+I(x)F(x)=0 \qquad F(x)=w_1F_1(x)+w_2F_2(x)$$

Another common type of differential equation is the *inhomogeneous* variant of 24:14:1, 24:14:6 $$\frac{d^2}{dx^2}f(x)-2B(x)\frac{d}{dx}f(x)+C(x)f(x)=R(x)$$

with a right-hand member, R(x), that is either a constant or a function of x. To solve this equation, first ignore the right-hand member and solve the corresponding homogeneous equation, 24:14:1. Then, knowing f_1 and f_2, the complete solution of 24:14:6 is

24:14:7 $$f=f_1(x)\left[w_1-\int_{x_0}^{x}\frac{R(t)f_2(t)}{W(t)}dt\right]+f_2(x)\left[w_2+\int_{x_0}^{x}\frac{R(t)f_1(t)}{W(t)}dt\right]$$

with x_0 arbitrary, and where

24:14:8 $$W(t)=f_1(t)\frac{d}{dt}f_2(t)-f_2(t)\frac{d}{dt}f_1(t)=f_1^2(t)\frac{d}{dt}\frac{f_2(t)}{f_1(t)}$$

Thus, at least in principle, if the solution of 24:14:1 can be found, so can that of 24:14:6. The integrals in 24:14:7 are known as *particular integrals*. The quantity W(t) is the *Wronskian* (Josef-Maria Hoëné de Wronski, 1778 – 1853, Polish, then French, mathematician and philosopher) of the two functions f_1 and f_2. These two solutions are linearly independent only if the Wronskian is nonzero. See Section 26:3 for a simple example.

24:15 RELATED TOPICS: Gauss-Hermite integration and other Gaussian quadratures

Suppose one needs to integrate a function of the form f(t)exp($-t^2$) numerically, between limits of $\pm\infty$. A powerful way of doing this uses the orthogonality properties [Section 21:14] of Hermite polynomials to generate the approximation

24:15:1 $$\int_{-\infty}^{\infty}f(t)\exp(-t^2)\,dt\approx\frac{2^{n-1}n!\sqrt{\pi}}{n^2}\sum_{j=1}^{n}\frac{f(r_j)}{H_{n-1}^2(r_j)}\qquad n=2,3,4,\cdots$$

where the set $r_1, r_2, \cdots, r_n$ includes all the zeros of the Hermite $H_n(t)$ polynomial. Choosing $n=4$ as a simple example, and using result 24:7:2, one finds

24:15:2 $$\int_{-\infty}^{\infty}f(t)\exp(-t^2)dt\approx12\sqrt{\pi}\left\{\frac{f\left(-\sqrt{\frac{3}{2}+\sqrt{\frac{3}{2}}}\right)}{H_3^2\left(-\sqrt{\frac{3}{2}+\sqrt{\frac{3}{2}}}\right)}+\frac{f\left(-\sqrt{\frac{3}{2}-\sqrt{\frac{3}{2}}}\right)}{H_3^2\left(-\sqrt{\frac{3}{2}-\sqrt{\frac{3}{2}}}\right)}+\frac{f\left(\sqrt{\frac{3}{2}-\sqrt{\frac{3}{2}}}\right)}{H_3^2\left(\sqrt{\frac{3}{2}-\sqrt{\frac{3}{2}}}\right)}+\frac{f\left(\sqrt{\frac{3}{2}+\sqrt{\frac{3}{2}}}\right)}{H_3^2\left(\sqrt{\frac{3}{2}+\sqrt{\frac{3}{2}}}\right)}\right\}$$

More usually, one would need to precalculate the zeros [by the Newton-Raphson method of Section 52:15, for example] or use tabulated values [such as those in Chapter 25 of Abramowitz and Stegun].

The Gauss-Hermite integration formula 24:15:2 is just one of a large number of integration schemes known collectively as *Gauss integration*, or *Gaussian quadrature*. They share the characteristic of being weighted sums of the values of the function f evaluated at the zeros of an orthogonal polynomial, the weights often being inversely proportional to the square of the polynomial of one order lower. Some other examples are *Gauss-Laguerre integration*

24:15:3 $$\int_{0}^{\infty}f(t)\exp(-t)dt\approx\frac{1}{n^2}\sum_{j=1}^{n}\frac{r_j\,f(r_j)}{L_{n-1}^2(r_j)}\qquad n=2,3,4,\cdots$$

and *Gauss-Legendre integration*

24:15:4
$$\int_{-1}^{1} f(t)\,dt \approx \frac{2}{n^2}\sum_{j=1}^{n}\frac{(1-r_j^2)\,f(r_j)}{P_{n-1}^2(r_j)} \qquad n = 2,3,4,\cdots$$

The weights are uniform for *Gauss-Chebyshev integration*

24:15:5
$$\int_{-1}^{1} \frac{f(t)}{\sqrt{1-t^2}}\,dt \approx \frac{\pi}{n}\sum_{j=1}^{n} f(r_j) \qquad n = 2,3,4,\cdots$$

in which r_j is the jth zero of the Chebyshev $T_n(t)$ polynomial.

The derivation of the Gaussian quadrature formulas [Matthews and Walker, Section 13-2] treats the f function as a polynomial in t. The closer that is to being the truth, and the larger n is, the better the approximation. If f is, in fact, a polynomial of degree $2n-1$ or less, the formulas are exact.